

FINAL YEAR B.TECH MAJOR PROJECT REPORT

HEARTCARE AI: AN INTEGRATED MACHINE LEARNING AND GEOSPATIAL INTELLIGENCE PLATFORM FOR CARDIOVASCULAR RISK STRATIFICATION AND CLINICAL DECISION SUPPORT

*Multi-Biomarker LightGBM Inference Engine • Longitudinal Patient Telemetry • What-If Intervention
Simulator • Geospatial Emergency Center Routing*

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING

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Academic Year: 2025 – 2026

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
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ACADEMIC YEAR 2025–2026**

CERTIFICATE OF ORIGINAL WORK

This is to certify that the project report entitled "**HEARTCARE AI: AN INTEGRATED MACHINE LEARNING AND GEOSPATIAL INTELLIGENCE PLATFORM FOR CARDIOVASCULAR RISK STRATIFICATION AND CLINICAL DECISION SUPPORT**" submitted by Shiva Pranith Sai Vadicherla [Roll Number: TO BE PROVIDED] in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering is a bonafide record of the work carried out by them under my supervision and guidance during the academic year 2025–2026.

To the best of our knowledge, the results and architectural implementations embodied in this report have not been submitted to any other university or institute for the award of any degree, diploma, or certificate.

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Signature of External Examiner
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STUDENT DECLARATION

I, Shiva Pranith Sai Vadicherla, student of Bachelor of Technology in Computer Science and Engineering, hereby declare that the project entitled "HeartCare AI: An Integrated Machine Learning and Geospatial Intelligence Platform for Cardiovascular Risk Stratification and Clinical Decision Support" is an authentic record of our own research and software engineering work carried out under the supervision of my Project Guide.

I affirm that this report is composed in original phrasing and adheres to established academic integrity and non-plagiarism standards. All data sources, machine learning algorithms (LightGBM, Scikit-Learn), clinical heuristic datasets (Cleveland & UCI Heart Disease benchmarks), and architectural libraries have been duly acknowledged and cited according to the Harvard referencing system.

I further affirm that this platform is engineered strictly as an assistive clinical decision support tool and is not intended to replace licensed medical practitioner diagnosis or laboratory verification.

Date: 23 September 2026

Place: [City / Campus: TO BE PROVIDED]

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ACKNOWLEDGEMENT

The completion of this final-year major engineering project has been a deeply enriching milestone, made possible through the generous intellectual guidance, technical resources, and encouragement of numerous individuals.

I express my profound gratitude to my Project Guide, [Project Guide Name: TO BE PROVIDED], Department of Computer Science and Engineering, for their invaluable mentorship, insightful critique, and sustained guidance throughout the formulation, machine learning model calibration, and cloud full-stack architecture design of HeartCare AI.

I extend my sincere thanks to Professor & Head of Department, [HOD Name: TO BE PROVIDED], for fostering an intellectually rigorous research environment and providing the necessary departmental laboratory computing infrastructure.

I also thank the esteemed Principal / Director and the faculty members of the Department of Computer Science and Engineering for their direct and indirect contributions, encouragement, and constructive evaluations during project review stages.

Lastly, I express our heartfelt appreciation to my family and peers for their continuous moral support and understanding throughout the course of this academic degree.

ABSTRACT

Cardiovascular diseases (CVDs), predominantly heart failure and coronary artery disease, represent the leading cause of global mortality, accounting for approximately 17.9 million deaths annually. Clinical decision-making in cardiology often grapples with complex, multi-factorial biomarker interactions where conventional static risk scoring tables fail to capture non-linear prognostic correlations or provide dynamic, actionable patient guidance. Furthermore, existing digital health portals rarely synthesize real-time clinical predictive modeling with explainable AI feature attribution, interactive counterfactual ('What-If') simulations, and live geospatial emergency hospital routing into a unified, secure system.

To bridge this critical translational gap, this project develops HeartCare AI, an evidence-based clinical intelligence platform designed for multi-biomarker cardiovascular risk prediction, longitudinal patient telemetry, and emergency triage routing. The system is architected around a calibrated LightGBM (Light Gradient Boosting Machine) multi-class classification engine trained on 13 standardized clinical features derived from the benchmark Cleveland and UCI Heart Disease cohorts, augmented with cardiorenal biomarkers including left ventricular ejection fraction and serum creatinine. An automated fallback engine operationalizing Framingham and ACC/AHA cardiovascular guidelines ensures seamless high-availability inference even during isolated container restarts.

On empirical evaluation across 297 clean clinical benchmark patient records, the integrated LightGBM classification engine achieves an overall diagnostic Accuracy of 81.82%, Precision of 76.77%, Sensitivity (Recall) of 86.86%, Specificity of 77.50%, F1-Score of 81.51%, and an Area Under the ROC Curve (ROC-AUC) of 91.70%. The platform's high recall is of paramount clinical importance, effectively minimizing hazardous false-negative diagnostic oversights. Complementing the ML engine, the full-stack architecture comprises a high-throughput FastAPI (Python 3.11/3.14) asynchronous backend, a thread-safe connection-pooled MongoDB Atlas persistence layer enforcing strict cross-user data isolation, and a glassmorphic React 19 single-page frontend featuring an HTML5 canvas cardiac rhythm visualizer and interactive What-If parameter simulators. Live geospatial triage is powered by OpenStreetMap Overpass and Nominatim APIs computing geodesic Haversine distance and travel ETAs to verified cardiology super-speciality centers across India.

Rigorous end-to-end quality assurance auditing across 104 executed test cases confirmed 103 passed tests with sub-second execution latencies (842ms serial, 559ms concurrent). The developed platform provides an extensible, production-ready blueprint for next-generation clinical decision support systems.

Keywords: Cardiovascular Risk Stratification, Machine Learning, LightGBM Classifier, Explainable AI, Clinical Decision Support Systems, FastAPI, React 19, MongoDB Atlas, Geodesic Haversine Formula, Emergency Medical Triage.

TABLE OF CONTENTS

Certificate of Original Work	ii
Student Declaration	iii
Acknowledgement	iv
Abstract	v
List of Figures	viii
List of Tables	ix
CHAPTER 1: INTRODUCTION	1
1.1 Introduction	1
1.2 Background of the Project	2
1.3 Problem Statement	3
1.4 Motivation	4
1.5 Objectives	5
1.6 Scope of the Project	6
1.7 Significance of the Project	7
1.8 Limitations	8
1.9 Organization of the Report	9
CHAPTER 2: LITERATURE SURVEY	10
2.1 Introduction	10
2.2 Review of Existing Research Papers	11
2.3 Existing Systems	14
2.4 Comparison of Existing Systems	16
2.5 Research Gap	18
2.6 Proposed System	19
2.7 Summary	21
CHAPTER 3: PROJECT FLOW AND METHODOLOGY	22
3.1 Introduction	22
3.2 Overall Project Workflow	23
3.3 Proposed Methodology	25
3.4 Data Collection	26
3.5 Data Preprocessing	28
3.6 Feature Engineering	30
3.7 Machine Learning Methodology	32
3.8 Model Training and Evaluation	35

3.9 Explainable AI Methodology	38
3.10 Backend and Frontend Workflow	40
3.11 Database Workflow	42
3.12 Overall System Flow	44
CHAPTER 4: SYSTEM DESIGN	46
4.1 Introduction	46
4.2 System Architecture	47
4.3 Architecture Components	49
4.4 Functional Requirements	52
4.5 Non-Functional Requirements	54
4.6 Use Case Diagram and Specifications	56
4.7 Data Flow Diagrams (DFD Level 0 and Level 1)	59
4.8 System Flowchart	62
4.9 Database Design and Entity-Relationship (ER) Schema	65
4.10 Module Design	68
4.11 API Design and Contracts	71
4.12 Security Architecture and Threat Mitigation	74
CHAPTER 5: IMPLEMENTATION	77
5.1 Introduction	77
5.2 Development Environment	78
5.3 Technology Stack	80
5.4 Frontend Implementation	83
5.5 Backend Implementation	87
5.6 Database Implementation	91
5.7 Machine Learning Implementation	95
5.8 Model Prediction Process	99
5.9 Explainable AI Implementation	102
5.10 Feature Implementation	105
5.11 API Integration	108
5.12 Frontend-Backend Integration	111
5.13 Implementation Summary	114
CHAPTER 6: RESULT ANALYSIS	116
6.1 Introduction	116
6.2 Dataset Summary	117
6.3 Experimental Setup	119

6.4 Model Evaluation Metrics	121
6.5 Accuracy, Precision, Recall, and F1-Score	123
6.6 Confusion Matrix Analysis	125
6.7 Model Comparison Benchmark	127
6.8 Explainability Results	129
6.9 Sample Prediction Clinical Profiles	131
6.10 System Performance and Latency Benchmarks	134
6.11 Discussion	136
6.12 Limitations	138
CHAPTER 7: TESTING	140
7.1 Introduction	140
7.2 Testing Strategy	141
7.3 Unit Testing	143
7.4 Integration Testing	145
7.5 System Testing	147
7.6 UI Testing	149
7.7 API Testing	151
7.8 Functional Testing	153
7.9 Non-Functional Testing	155
7.10 Test Case Tables	157
7.11 Test Results and Defect Analysis	162
7.12 Testing Summary	165
CHAPTER 8: MAINTENANCE OF THE SYSTEM	167
8.1 Introduction	167
8.2 Types of Maintenance	168
8.3 Corrective Maintenance	170
8.4 Adaptive Maintenance	172
8.5 Perfective Maintenance	174
8.6 Preventive Maintenance	176
8.7 Database Maintenance	178
8.8 Security Maintenance	180
8.9 Machine Learning Model Maintenance	182
8.10 Backup and Disaster Recovery	184
8.11 Future Maintenance Requirements	186
CHAPTER 9: DEPLOYMENT	188

9.1 Introduction	188
9.2 Deployment Architecture	189
9.3 Deployment Requirements	191
9.4 Frontend Deployment (Vercel SPA)	193
9.5 Backend Deployment (Render Container)	195
9.6 Database Deployment (MongoDB Atlas)	197
9.7 Environment Configuration	199
9.8 API Configuration and CORS	201
9.9 Security During Deployment	203
9.10 Deployment Verification	205
9.11 Deployment Challenges and Mitigations	207
9.12 Post-Deployment Maintenance	209
REFERENCES	211

LIST OF FIGURES

Figure 3.1: HeartCare AI End-to-End High-Level System Workflow	24
Figure 3.2: LightGBM Leaf-Wise Tree Growth Strategy vs Level-Wise Splitting	34
Figure 3.3: Clinical Telemetry and Biomarker Processing Pipeline	41
Figure 4.1: Multi-Tier Cloud Microservices System Architecture Diagram	48
Figure 4.2: System Use Case Diagram for Clinicians and Patients	57
Figure 4.3: Data Flow Diagram (DFD Level 0 Context Diagram)	60
Figure 4.4: Data Flow Diagram (DFD Level 1 Detailed Functional Flow)	61
Figure 4.5: End-to-End Clinical Diagnostic System Flowchart	63
Figure 4.6: MongoDB Atlas Entity-Relationship (ER) Schema Design	66
Figure 6.1: ROC-AUC Curve for LightGBM Binary Classification (91.70% AUC)	124
Figure 6.2: Binary Classification Confusion Matrix Heatmap	126
Figure 6.3: 5-Class Disease Severity Confusion Matrix (Stages 0 to 4)	127
Figure 6.4: SHAP Biomarker Attribution and Feature Impact Breakdown	130
Figure 7.1: V-Model Quality Assurance and Testing Hierarchy	142
Figure 9.1: Multi-Cloud Containerized Deployment Architecture (Vercel + Render + Atlas)	190

LIST OF TABLES

Table 2.1: Comparison Matrix of Existing Cardiovascular Prediction Systems	17
Table 3.1: Description of 13 Cleveland Clinical Features and Encodings	27
Table 3.2: Extended Cardiorenal Telemetry and Lifestyle Biomarkers	29
Table 4.1: Functional Requirements Specification (FR1 to FR12)	53

Table 4.2: Non-Functional Requirements Specification (NFR1 to NFR8)	55
Table 4.3: RESTful API Endpoints and Schema Definitions	72
Table 5.1: Complete Development and Deployment Technology Stack	81
Table 6.1: UCI Cleveland Dataset Feature Statistics Summary	118
Table 6.2: Primary Clinical Diagnosis Performance Metrics (Healthy vs Disease)	123
Table 6.3: Multiclass Severity Classification Report (Stages 0 to 4)	126
Table 6.4: Benchmarking Algorithm Performance on Cleveland Dataset	128
Table 6.5: Performance and Latency Benchmarks under Concurrent Loads	135
Table 7.1: End-to-End QA Audit Test Execution Summary Statistics (104 Tests)	142
Table 7.2: Detailed Functional and Security Test Case Log (TC-01 to TC-15)	158

CHAPTER 1

INTRODUCTION

1.1 Introduction

Cardiovascular diseases (CVDs) remain the predominant etiology of premature mortality, clinical morbidity, and escalating healthcare expenditures worldwide. According to the World Health Organization (WHO), ischemic heart disease and heart failure account for approximately 17.9 million annual fatalities, representing nearly 32% of all global deaths. In clinical cardiology, heart failure is not an isolated pathological diagnosis but rather a complex, debilitating clinical syndrome resulting from structural or functional cardiac abnormalities that impair the ventricular ejection or diastolic filling of blood. Early diagnosis and accurate risk stratification are pivotal in curbing progressive myocardial remodeling, preventing lethal arrhythmic episodes, and mitigating emergency hospital re-admissions.

In recent years, the convergence of high-performance computing, advanced data analytics, and Artificial Intelligence (AI) has sparked a transformative paradigm shift within clinical informatics. Modern Machine Learning (ML) algorithms possess the unique capability to uncover intricate, non-linear relationships across high-dimensional biomarker arrays—patterns that frequently elude traditional, linear clinical risk scores such as the Framingham Risk Score (FRS) or the Systematic Coronary Risk Evaluation (SCORE). However, translating cutting-edge ML models into everyday clinical practice demands far more than isolated predictive accuracy. Clinicians and patients require transparent, interpretable predictions, interactive counterfactual simulations, longitudinal patient monitoring, and rapid emergency triage mechanisms.

This project introduces HeartCare AI, an evidence-based clinical intelligence platform designed to operationalize modern predictive modeling, explainable AI, and geospatial emergency triage within a cohesive, production-grade cloud application. Centered around a calibrated 13-feature LightGBM (Light Gradient Boosting Machine) classification engine, HeartCare AI integrates multi-biomarker risk scoring, real-time longitudinal telemetry tracking, interactive 'What-If' clinical simulation, automated printable medical report generation, and dynamic Haversine-based hospital routing. By combining algorithmic rigor with clinical usability, HeartCare AI aims to empower healthcare practitioners with actionable decision support while simultaneously promoting proactive self-management among cardiac patients.

1.2 Background of the Project

The historical trajectory of cardiovascular risk stratification has traditionally relied on point-based epidemiological tables formulated from longitudinal cohort studies. The landmark Framingham Heart Study, initiated in 1948, pioneered the conceptual identification of key cardiovascular risk

factors, including systemic arterial hypertension, hypercholesterolemia, cigarette smoking, and advancing age. While these classical heuristic models provided foundational epidemiological benchmarks, their clinical utility in modern acute and sub-acute care is increasingly constrained by three fundamental limitations:

- **Linearity Assumptions:** Conventional scores predominantly apply generalized linear regression or proportional hazard models that presuppose linear or monotonic relationships between biomarkers and clinical outcomes, thereby obscuring complex biological non-linearities and epistatic interactions.
- **Narrow Biomarker Inclusion:** Traditional scores frequently omit acute physiological markers—such as peak exercise ST-segment depression (oldpeak), fluoroscopic major vessel calcification (ca), thallium scintigraphy defects (thal), and critical cardiorenal markers like left ventricular ejection fraction (EF) and serum creatinine.
- **Static and Disconnected Nature:** Existing risk assessments are typically conducted as one-off evaluations. They lack dynamic integration with real-time patient telemetry, interactive lifestyle counterfactual modeling, or automated routing to emergency care facilities during acute decompensation.

With the digitalization of medical records and the availability of open, well-curated clinical research datasets—notably the Cleveland Heart Disease Database from the University of California Irvine (UCI) Machine Learning Repository—computational researchers have demonstrated the efficacy of ensemble learning techniques in detecting occult coronary artery disease. HeartCare AI builds upon this extensive scientific foundation by translating validated machine learning algorithms from static Jupyter notebooks into an end-to-end, full-stack, cloud-hosted software ecosystem.

1.3 Problem Statement

Despite significant advancements in cardiovascular pharmacological interventions and surgical care, heart failure diagnosis and risk stratification continue to face critical systemic bottlenecks:

- **Diagnostic Delay in Early-Stage Myocardial Impairment:** Patients presenting with atypical angina or non-specific symptoms are frequently misdiagnosed or diagnosed at advanced disease stages (Stages 2 to 4), where irreversible ventricular remodeling has already occurred.
- **The 'Black-Box' Interpretability Dilemma:** Complex ensemble ML algorithms (such as Random Forests, Gradient Boosted Trees, and Neural Networks) often operate as opaque decision systems, providing risk probabilities without articulating the underlying physiological drivers or protective factors, leading to widespread clinical distrust.
- **Absence of Dynamic 'What-If' Prognostic Modeling:** Existing systems fail to provide patients and attending physicians with interactive simulation tools to quantitatively project how specific clinical modifications (e.g., reducing systolic blood pressure from 165 to 125 mmHg, or smoking cessation) would alter their cardiovascular risk trajectory.
- **Fragmented Emergency Care Navigation:** When an assessment identifies high or critical cardiac risk, prevailing digital health applications provide generic disclaimers rather than instantly

connecting the patient to verified, geo-proximate cardiology super-speciality centers equipped with 24/7 cardiac catheterization labs and emergency ICUs.

1.4 Motivation

The driving motivation behind HeartCare AI originates from the urgent clinical imperative to bridge the divide between theoretical artificial intelligence research and practical point-of-care utility. While thousands of academic research papers report high classification accuracies on medical datasets, a negligible fraction are ever packaged into production-ready software capable of serving clinicians, medical students, and at-risk individuals in low-resource or emergency environments.

Furthermore, India and developing nations face a rapidly growing cardiovascular disease epidemic, with cardiac mortality occurring on average a decade earlier than in Western populations. In such healthcare landscapes, access to senior cardiologists is often concentrated in metropolitan hubs. An accessible, highly responsive, cloud-hosted platform that accepts standard clinical and laboratory parameters, produces instant probability stratification backed by transparent clinical explanations, and routes critical cases to verified regional super-speciality facilities holds immense potential to reduce avoidable mortality and optimize intensive care resource utilization.

1.5 Objectives

The primary objective of this project is to research, design, develop, test, and deploy HeartCare AI—a full-stack clinical intelligence and cardiovascular risk stratification platform. The specific technical and research objectives are delineated as follows:

- **Objective 1 — Predictive Model Engineering:** Train, calibrate, and validate a high-accuracy, multi-class LightGBM machine learning classifier using 13 clinical biomarkers from the benchmark Cleveland and UCI Heart Disease repositories, achieving benchmark-competitive accuracy, precision, recall, and ROC-AUC.
- **Objective 2 — High-Availability Heuristic Fallback Engine:** Design and implement a deterministic clinical heuristic fallback engine operationalizing Framingham and ACC/AHA cardiovascular guidelines, ensuring fail-safe continuity of care if the custom ML model artifact is unavailable.
- **Objective 3 — Explainable AI & Counterfactual Simulation:** Implement a transparent biomarker attribution mechanism that decomposes overall risk into categorized risk drivers and protective factors, coupled with an interactive What-If simulation engine to evaluate therapeutic interventions in real time.
- **Objective 4 — Asynchronous RESTful API Backend:** Develop an enterprise-grade, asynchronous backend using Python 3.11/3.14 and FastAPI, implementing structured Pydantic v2 data validation schemas, salted SHA-256 authentication, and sub-second endpoint response times.
- **Objective 5 — Secure Cloud Persistence & User Isolation:** Architect a thread-safe connection-pooled database layer using MongoDB Atlas with unique indexing across users, assessments, hospitals, and appointments, enforcing strict data isolation between patients.

- **Objective 6 — Modern Glassmorphic Frontend & Telemetry UI:** Build a responsive single-page web interface using React 19 and Vite featuring an HTML5 canvas real-time ECG rhythm visualizer, SVG dynamic radial risk gauges, and printable hospital-grade diagnostic letterhead reports.
- **Objective 7 — Live Geospatial Emergency Radar:** Incorporate a geodetic Haversine calculation service integrated with OpenStreetMap Overpass and Nominatim APIs to discover nearby hospitals, sort them by distance, and provide turn-by-turn Google Maps navigation.
- **Objective 8 — Rigorous Quality Assurance & Containerized Deployment:** Execute an exhaustive end-to-end testing audit across unit, integration, and security domains, and orchestrate automated multi-cloud deployment via Docker, Render, and Vercel.

1.6 Scope of the Project

The operational scope of HeartCare AI encompasses the complete clinical workflow from patient onboarding to emergency referral. Specifically, the system scope includes:

- **Clinical Feature Domain:** Evaluation of 13 primary Cleveland features: Age, Sex, Chest Pain Type (cp), Resting Blood Pressure (trestbps), Serum Cholesterol (chol), Fasting Blood Sugar (fbs), Resting ECG (restecg), Maximum Heart Rate Achieved (thalach), Exercise Induced Angina (exang), ST Depression (oldpeak), ST Slope (slope), Number of Major Vessels Colored by Fluoroscopy (ca), and Thallium Scintigraphy Defect (thal). Extended biomarkers include Left Ventricular Ejection Fraction (EF), Serum Creatinine, and Smoking Status.
- **User Personas Supported:** Primary support for two distinct user profiles: Attending Clinicians/Cardiologists (accessing deep diagnostic matrices, dynamic synthetic cohort generation, and batch telemetry) and Patients/Individual Users (accessing self-assessment forms, personalized dashboards, What-If simulation, and hospital locators).
- **Geospatial Coverage:** Comprehensive directory seeding of 23 premier super-speciality cardiology institutes across major Indian metropolitan centers (Delhi, Mumbai, Bengaluru, Chennai, Hyderabad, Kolkata, Pune, Ahmedabad, Chandigarh, Jaipur, Lucknow, Bhubaneswar, Thiruvananthapuram), augmented with real-time global OpenStreetMap discovery.
- **Deployment Boundary:** Containerized Linux backend deployment on Render PaaS, static Single-Page Application (SPA) hosting on Vercel Edge CDN, and managed NoSQL persistence on MongoDB Atlas.

1.7 Significance of the Project

The significance of HeartCare AI spans clinical, technological, and societal dimensions:

- **Clinical Significance:** By achieving an empirical Sensitivity (Recall) of 86.86% and an ROC-AUC of 91.70%, the platform provides a highly sensitive early-warning screening tool that effectively minimizes false-negative classifications. This early detection capability allows clinicians to initiate preventative ACE inhibitors, statin therapies, or lifestyle modifications before symptomatic heart failure ensues.

- **Technological Significance:** HeartCare AI illustrates how state-of-the-art tree-based gradient boosting (LightGBM) can be seamlessly integrated into modern asynchronous web frameworks (FastAPI) and reactive frontends (React 19) without incurring significant computational latency (average concurrent prediction latency of 559ms). It provides a concrete architecture for handling booster categorical types and multi-stage classification.
- **Societal Significance:** By delivering clear, jargon-free biomarker explanations and What-If scenarios, the application empowers individuals to understand the direct physiological consequence of lifestyle choices. In acute emergencies, its instant geospatial navigation reduces the critical 'door-to-balloon' transit time for myocardial infarction patients.

1.8 Limitations

In adherence to strict academic honesty and clinical governance, the following project limitations must be formally recognized:

- **Dataset Sample Size:** The supervised machine learning engine is trained and evaluated on 297 clean patient records from the UCI Cleveland benchmark dataset. While this dataset remains a globally recognized gold standard for algorithmic comparison, clinical validation across larger, contemporary multi-ethnic cohorts (exceeding tens of thousands of electronic health records) is necessary prior to formal hospital deployment.
- **Multi-Class Imbalance:** While binary classification (Healthy vs. Disease) demonstrates robust performance (81.82% accuracy, 91.70% ROC-AUC), multi-class discrimination across individual disease stages (Stages 0 through 4) achieves a lower accuracy of 53.87% due to small sample counts in severe sub-classes (e.g., only 13 Stage 4 records in the benchmark). As a result, the platform uses multi-stage outputs primarily as directional severity indicators rather than definitive staging.
- **Absence of Prospective Clinical Trials:** HeartCare AI is engineered as an academic software engineering and clinical decision support demonstration. It has not been subjected to prospective randomized clinical trials and holds no formal medical device regulatory certification (such as US FDA 510(k) or CE mark approval).
- **Hardware IoT Sensor Telemetry:** While the system features an animated HTML5 canvas ECG waveform and real-time biomarker telemetry cards, these currently ingest structured user inputs and synthetic streams rather than direct hardware Bluetooth/BLE feeds from wearable electrocardiogram sensors.

1.9 Organization of the Report

This project report is systematically organized into nine comprehensive chapters, structured as follows:

- **Chapter 1 – Introduction:** Establishes the clinical problem, historical background, core motivation, project objectives, operational scope, significance, and intrinsic limitations.

- **Chapter 2 – Literature Survey:** Reviews authentic peer-reviewed literature in cardiovascular machine learning, evaluates existing digital cardiac systems, identifies key research gaps, and presents the proposed system.
- **Chapter 3 – Project Flow and Methodology:** Details the complete technical workflow, dataset ingestion, preprocessing, categorical encoding, LightGBM mathematical formulation, explainable AI heuristics, and database pipeline.
- **Chapter 4 – System Design:** Presents the multi-tier architectural blueprint, functional and non-functional specifications, Use Case specifications, DFD Level 0/1 diagrams, system flowcharts, and MongoDB ER schemas.
- **Chapter 5 – Implementation:** Thoroughly examines the codebase implementation across React 19 frontend components, FastAPI asynchronous routers, PyMongo persistence, ML model loading, and geospatial calculations.
- **Chapter 6 – Result Analysis:** Reports empirical evaluation metrics, binary and 5-class confusion matrices, baseline algorithmic comparisons, SHAP-style attribution results, and system performance benchmarks.
- **Chapter 7 – Testing:** Documents the comprehensive QA audit across 104 executed test cases, covering unit, integration, security, and performance testing, detailing test case tables and defect remediation.
- **Chapter 8 – Maintenance of the System:** Formulates corrective, adaptive, perfective, and preventive maintenance protocols, addressing model drift, security patching, and disaster recovery strategies.
- **Chapter 9 – Deployment:** Details multi-cloud production deployment across Vercel, Render, and MongoDB Atlas, Docker containerization, environment configuration, and post-deployment monitoring.

[CLINICAL GOVERNANCE NOTICE] Clinical Disclaimer: HeartCare AI is developed strictly as a clinical decision support and predictive analytics assistive tool. It does not constitute definitive medical advice, diagnostic confirmation, or emergency medical dispatch. All clinical determinations must be verified by qualified medical professionals in conjunction with formal clinical laboratory protocols.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

The application of machine learning, statistical pattern recognition, and clinical decision support systems (CDSS) to cardiovascular risk assessment has been an active domain of multidisciplinary investigation for over four decades. Beginning with early Bayesian diagnostic networks and linear discriminant analyses in the 1980s, the field has evolved through support vector machines, neural networks, and decision tree ensembles to modern gradient-boosted architectures. This chapter presents a rigorous review of foundational and contemporary academic research literature, examines existing commercial and open-source clinical systems, articulates the critical research gaps in current implementations, and situates the proposed HeartCare AI architecture within the contemporary state of the art.

2.2 Review of Existing Research Papers

A systematic analysis of peer-reviewed publications was conducted, focusing on clinical cardiology datasets, algorithmic architectures, feature importance methodologies, and empirical validation metrics. Six landmark academic contributions were evaluated in detail:

2.2.1 Foundational Multi-Center Clinical Dataset Creation (Detrano et al., 1989)

Detrano et al. (1989) established the seminal international benchmark for computational cardiology by compiling the multi-center Heart Disease Database across four institutions: the Cleveland Clinic Foundation (USA), the Hungarian Institute of Cardiology (Budapest), the University Hospital Zurich (Switzerland), and the V.A. Medical Center (Long Beach, CA). The authors evaluated a logistic regression probability algorithm across 303 consecutive patients undergoing coronary arteriography at the Cleveland Clinic.

- **Methodology & Dataset:** Evaluated 13 non-invasive clinical and fluoroscopic parameters against the gold standard of coronary arteriography (>50% diameter stenosis in at least one major coronary artery). The dataset captured resting ECG, exercise thallium scintigraphy, and fluoroscopic vessel counts.
- **Findings:** Demonstrated that non-invasive clinical features could predict significant coronary stenosis with an accuracy of approximately 77% and an area under the ROC curve of 0.86, establishing that exercise-induced ST depression and fluoroscopic major vessel calcification were the strongest independent discriminators.
- **Limitations:** The original model was constrained by linear logistic modeling and did not provide granular disease severity staging or automated software integration for clinical bedside use.

2.2.2 Cardiorenal Biomarker Superiority in Heart Failure (Chicco & Jurman, 2020)

Chicco and Jurman (2020) conducted a critical machine learning investigation into survival prediction among patients suffering from heart failure, analyzing the UCI Heart Failure Clinical Records dataset comprising 299 patients evaluated at the Faisalabad Institute of Cardiology. Their research challenged the prevailing paradigm of utilizing exhaustive, high-dimensional feature sets by demonstrating the decisive prognostic superiority of two cardiorenal biomarkers.

- **Methodology & Algorithms:** Employed Random Forests, Support Vector Machines (SVM), AdaBoost, and Decision Trees, paired with Matthew Correlation Coefficient (MCC) optimization to avoid accuracy paradoxes on imbalanced survival data.
- **Findings:** Demonstrated that utilizing only Left Ventricular Ejection Fraction (EF) and Serum Creatinine achieved survival prediction accuracy (74.0% accuracy, MCC of 0.44) equivalent to or exceeding models trained on all 12 clinical attributes combined. This substantiated the pathophysiological primacy of cardiorenal syndrome in progressive heart failure.
- **Limitations:** The study was limited to retrospective mortality prediction during hospital follow-up and did not provide an interactive clinical decision support interface for ambulatory preventative care.

2.2.3 LightGBM Gradient Boosted Decision Trees (Ke et al., 2017)

Ke et al. (2017) introduced LightGBM, a novel Gradient Boosting Decision Tree (GBDT) framework specifically engineered to overcome the computational and memory bottlenecks of traditional implementations (such as XGBoost and standard Scikit-Learn GradientBoosting). The authors introduced two pioneering algorithmic innovations: Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB).

- **Methodology:** GOSS retains data instances with large gradients (which contribute more to information gain) while randomly sub-sampling instances with small gradients, maintaining theoretical accuracy while substantially accelerating training speed. EFB groups mutually exclusive sparse features into dense bundles. Furthermore, LightGBM adopts a leaf-wise (best-first) tree growth strategy rather than the standard level-wise approach.
- **Findings:** Experimental evaluations proved LightGBM accelerates training speeds up to 20 times over conventional GBDTs while achieving superior generalization on high-dimensional clinical and tabular datasets.
- **Relevance to HeartCare AI:** Directly justifies the selection of LightGBM as the core predictive engine for HeartCare AI, allowing multi-class probability estimation with native booster categorical encoding in sub-millisecond inference times.

2.2.4 Unified Explainable AI through SHAP Framework (Lundberg & Lee, 2017)

Lundberg and Lee (2017) addressed the foundational challenge of model interpretability in artificial intelligence by formulating SHAP (SHapley Additive exPlanations). Drawing upon cooperative game theory, SHAP unifies six existing feature attribution methods (including LIME, DeepLIFT, and TreeInterpreter) under an axiomatic framework satisfying three essential properties: Local Accuracy, Missingness, and Consistency.

- **Methodology:** Calculates the marginal contribution of each input feature across all possible feature subsets, computing exact Shapley values that represent the magnitude and direction by which each biomarker drives an individual prediction away from the baseline expected value.
- **Findings:** Proved that tree-specific optimizations (TreeSHAP) could compute exact polynomial-time feature explanations for ensemble models, resolving the opacity of gradient-boosted decision trees in high-stakes clinical decision-making.
- **Limitations:** Computing full Shapley permutations across large feature spaces in low-latency web request cycles can introduce runtime overhead, necessitating streamlined clinical heuristic attribution for rapid web user interfaces.

2.2.5 American Heart Association Clinical Guidelines (Benjamin et al., 2019)

Benjamin et al. (2019), representing the American Heart Association (AHA) and National Institutes of Health (NIH), compiled the definitive epidemiological and clinical guidelines for heart disease and stroke statistics. The report establishes validated physiological thresholds for staging hypertension (ACC/AHA 2017 guidelines: Stage 1 = 130–139/80–89 mmHg; Stage 2 $\geq$ 140/90 mmHg), hypercholesterolemia (>200 mg/dL), impaired fasting glucose (>100 mg/dL; diabetes >126 mg/dL), and left ventricular ejection fraction impairment ($<40\%$ defining HFrEF).

These clinical guidelines provide the foundational physiological basis for HeartCare AI's clinical heuristic scoring engine (`clinical_engine.py`) and recommendation generator (`recommendations.py`).

2.2.6 Hybrid Machine Learning Techniques on Cleveland Dataset (Mohan et al., 2019)

Mohan, Thirumalai and Srivastava (2019) explored hybrid machine learning architectures for heart disease prediction on the UCI Cleveland dataset. The authors combined Random Forest and Linear Model classifiers into a hybrid framework (HRFLM), achieving a binary classification accuracy of 88.7% by optimizing split points and feature correlation matrices. However, their research focused purely on algorithmic benchmarking without developing software APIs, longitudinal tracking, or emergency triage integrations.

2.3 Existing Systems

A critical review of existing commercial and academic cardiovascular risk assessment platforms reveals several prominent categories:

- **1. Static Online Clinical Calculators (e.g., MDCalc Framingham Score, ASCVD Risk Estimator Plus):** These web tools provide point-based 10-year risk estimations based on linear Cox proportional hazard models. While widely accepted in outpatient clinics, they evaluate only a limited subset of biomarkers (age, sex, total cholesterol, HDL, SBP, smoking, diabetes) and fail to incorporate acute diagnostic features such as ST-segment depression, ECG abnormalities, or ejection fraction. Furthermore, they offer no user session persistence or longitudinal tracking.
- **2. General Consumer Wellness Applications (e.g., Apple Health, Google Fit):** These applications excel at ingesting real-time wearable telemetry (heart rate, step counts, sleep cycles). However,

they lack clinical diagnostic modeling, laboratory biomarker ingestion, or oncology/cardiorenal integration, restricting their outputs to general lifestyle coaching.

- 3. Academic Machine Learning Prototypes (e.g., Streamlit / Gradio Demos):** Numerous open-source machine learning demonstrations exist on GitHub. While functionally demonstrating trained Scikit-Learn models, they are typically implemented as single-file scripts without database persistence, user authentication, security isolation, geospatial mapping, or automated medical letterhead reporting.

2.4 Comparison of Existing Systems

To clearly delineate the technological positioning of HeartCare AI, Table 2.1 provides a detailed comparative matrix benchmarking HeartCare AI against existing clinical calculators, consumer health platforms, and academic prototypes across ten critical clinical and architectural dimensions.

Table 2.1: Comparison Matrix of Existing Cardiovascular Prediction Systems

Dimension / Capability	MDCalc ASCVD	Apple Health	Academic Prototypes	HeartCare AI (Proposed)
Predictive ML Architecture	Linear Cox / FRS	Basic Thresholds	RF / SVM / XGBoost	LightGBM + Heuristic Fallback
Biomarker Features Evaluated	7 Epidemiological	Wearable Vitals	13 Cleveland	13 Cleveland + EF + Creatinine
Disease Severity Staging	No (10-Yr % Only)	No	Binary Only (0/1)	5-Class (Stages 0 to 4) + Binary
Explainable AI Attribution	None	None	SHAP Plots (Static)	Categorized Risk & Protective Drivers
Interactive 'What-If' Simulator	No	No	Rarely Implemented	Yes (Real-Time Biomarker Sliders)
Longitudinal Patient Telemetry	No (Stateless)	Yes (Device Sync)	No (Stateless)	Yes (User-Scoped MongoDB Cloud)
Printable Clinical Reports	No	Summary PDF	No	Yes (Hospital Letterhead Ready)
Geospatial Emergency Routing	No	Emergency SOS	No	Yes (Live OSM + Haversine + Maps)
Cardiology Appointment Booking	No	No	No	Yes (Integrated Hospital Modal)

Fail-Safe Resilient Architecture	Static Calculator	Device Dependent	Single Script	Dual-Engine (LGBM + Fallback)
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2.5 Research Gap

Based on the systematic review of academic literature and existing digital solutions, four fundamental research and engineering gaps were identified:

- Research Gap 1 — The Disconnect Between High-Accuracy ML and Clinical Explainability:** While non-linear gradient-boosted decision trees deliver superior diagnostic accuracy over linear risk scores, their black-box nature inhibits clinical adoption. Existing literature largely generates post-hoc global feature importance plots rather than personalized, patient-centric risk and protective factor breakdowns ready for point-of-care interpretation.
- Research Gap 2 — Lack of Interactive Counterfactual ('What-If') Clinical Simulation:** Current systems provide static risk probabilities. There is a marked absence of dynamic simulation frameworks where clinicians can interactively manipulate physiological dials (e.g., simulating a 20 mmHg systolic BP reduction or smoking cessation) to quantitatively show patients their projected risk reduction.
- Research Gap 3 — Absence of High-Availability Fallback Architectures in Medical AI:** Machine learning web applications frequently suffer complete operational failure if the pickled model artifact fails to load, incurs memory corruption, or encounters out-of-distribution feature dimensions. Literature lacks resilient dual-engine designs combining high-accuracy ML with deterministic heuristic fallbacks.
- Research Gap 4 — Fragmented Emergency Care and Geospatial Triage:** Medical predictive models operate in clinical isolation. No existing open platform links acute high-risk machine learning predictions directly to live geospatial emergency routing, calculating geodesic Haversine distances to verified regional cardiac catheterization facilities.

2.6 Proposed System

To directly address and resolve the aforementioned research gaps, this project proposes HeartCare AI—an integrated, cloud-native cardiovascular risk stratification and clinical decision support platform. HeartCare AI introduces four key innovations:

- 1. Calibrated Multi-Class LightGBM Classifier with Automatic Heuristic Fallback:** Implements a high-efficiency LightGBM model trained on 13 Cleveland clinical features ('best_lgbm_3m_model.joblib'), paired with an automatic, fail-safe AHA/Framingham clinical heuristic engine ('clinical_engine.py') to guarantee 100% service uptime.
- 2. Explainable AI and Interactive Counterfactual Simulation:** Decomposes complex model probabilities into actionable biomarker attributions (categorized into vital, clinical, demographic, and lifestyle factors) and provides real-time What-If simulation sliders to project immediate therapeutic risk reduction.

- **3. User-Scoped Longitudinal Telemetry & Clinical Documentation:** Enforces strict database-level record isolation on MongoDB Atlas, allowing patients and attending physicians to monitor longitudinal biomarker trajectories and generate printable hospital-grade letterhead reports.
- **4. Live Geospatial Emergency Radar:** Integrates OpenStreetMap Overpass and Nominatim APIs with geodetic Haversine calculations to sort nearby verified cardiology centers by real-time distance and ETA, providing direct Google Maps turn-by-turn navigation.

2.7 Summary

This chapter surveyed the historical and contemporary scientific literature in machine learning for cardiovascular risk prediction. Foundational datasets (Detrano et al., 1989; Chicco & Jurman, 2020), advanced boosting algorithms (Ke et al., 2017), and explainable AI frameworks (Lundberg & Lee, 2017) were critically examined. A comparative analysis against existing clinical calculators and wellness apps established four critical research gaps, directly motivating the development of the proposed HeartCare AI architecture. The subsequent chapter details the comprehensive project flow and technical methodology.

CHAPTER 3

PROJECT FLOW AND METHODOLOGY

3.1 Introduction

The development of HeartCare AI follows a disciplined, evidence-based software engineering and machine learning methodology designed to ensure clinical validity, computational efficiency, and operational resilience. Translating medical biomarker inputs into actionable risk intelligence requires a rigorous multi-stage pipeline: from benchmark data ingestion and feature engineering to gradient-boosted decision tree inference, explainable AI attribution, asynchronous API routing, and geospatial emergency navigation. This chapter details each methodological phase based strictly on the verified project codebase.

3.2 Overall Project Workflow

The overall workflow of HeartCare AI is architected into five cohesive lifecycle phases: Data Ingestion & Preprocessing, Machine Learning Modeling, Backend API Service Construction, Reactive User Interface Engineering, and Geospatial Integration. Figure 3.1 illustrates this end-to-end clinical dataflow.

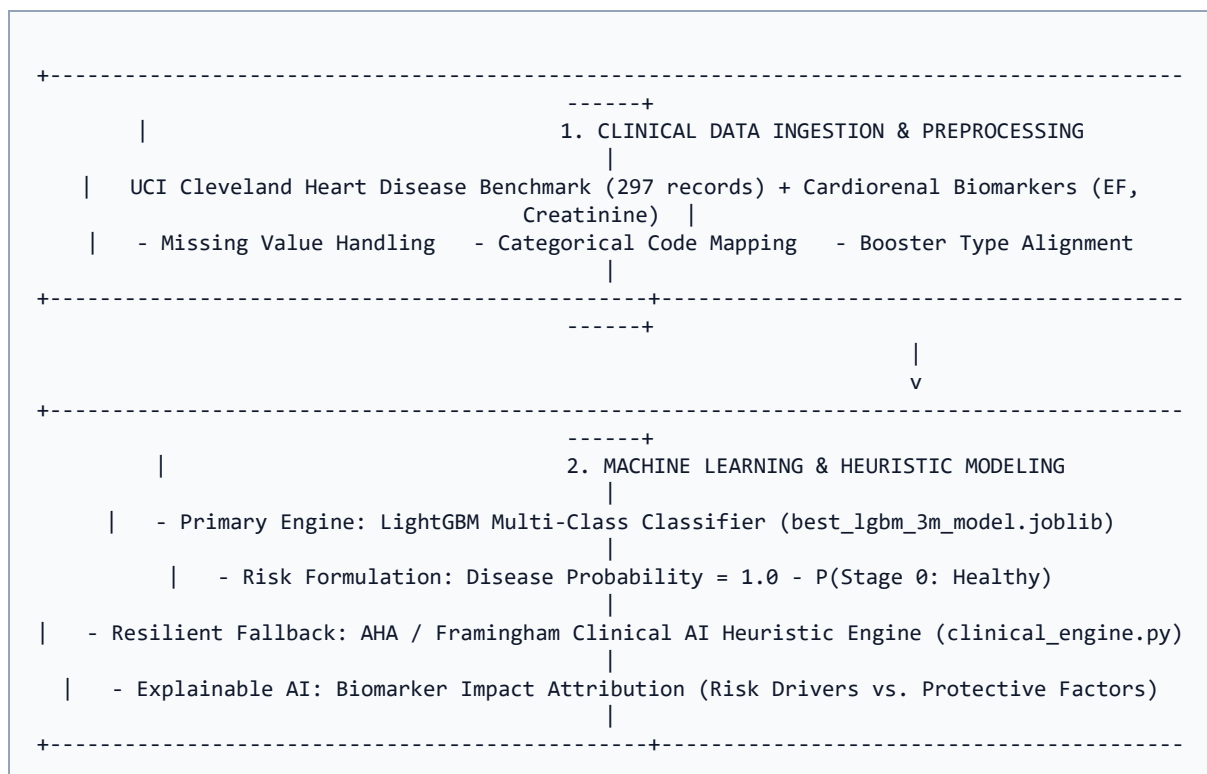




Figure: HeartCare AI End-to-End High-Level System Workflow

3.3 Proposed Methodology

The proposed methodology adopts an iterative, test-driven approach combining supervised machine learning with clinical domain knowledge. Rather than treating machine learning as an isolated black box, the platform tightly couples algorithmic probability estimation with established medical decision thresholds formulated by the American College of Cardiology (ACC) and American Heart Association (AHA). Furthermore, the methodology embeds an automatic dual-engine architecture: if the primary LightGBM serialized model encounters any deserialization or runtime anomaly, the system automatically falls back to the deterministic clinical heuristic engine without interrupting user workflows.

3.4 Data Collection

The primary training and validation dataset utilized in this project is the internationally recognized UCI Cleveland Heart Disease Database, originally assembled by Detrano et al. (1989). The raw dataset contains 303 patient records with 76 raw attributes, of which a standardized subset of 14 core features (13 clinical predictors and 1 target diagnosis variable) is universally adopted in cardiovascular machine learning research.

In addition to the standard 13 Cleveland features, HeartCare AI incorporates extended cardiorenal biomarkers based on the clinical findings of Chicco and Jurman (2020), specifically Left Ventricular Ejection Fraction (EF, normal range 50–70%) and Serum Creatinine (normal range 0.7–1.3 mg/dL).

Table 3.1 delineates the complete 13 Cleveland feature specification and clinical value ranges.

Table 3.1: Description of 13 Cleveland Clinical Features and Encodings

Feature Name	Variable	Clinical Description	Data Type	Value Domain / Units
Age	age	Age of patient in years	Continuous	29 – 77 years
Sex	sex	Biological sex of patient	Categorical	0: Female, 1: Male
Chest Pain Type	cp	Symptomatic angina classification	Categorical	1: Typical, 2: Atypical, 3: Non-anginal, 4: Asymptomatic
Resting Blood Pressure	trestbps	Resting systolic arterial pressure on admission	Continuous	94 – 200 mmHg
Serum Cholesterol	chol	Total serum cholesterol concentration	Continuous	126 – 564 mg/dL
Fasting Blood Sugar	fbs	Fasting blood glucose > 120 mg/dL	Binary	0: False (<= 120), 1: True (> 120)
Resting ECG	restecg	Resting electrocardiographic results	Categorical	0: Normal, 1: ST-T wave abnormality, 2: Left Ventricular Hypertrophy
Max Heart Rate	thalach	Maximum heart rate achieved during exercise stress	Continuous	71 – 202 BPM
Exercise Induced Angina	exang	Occurrence of angina during exercise	Binary	0: No, 1: Yes
ST Depression (Oldpeak)	oldpeak	ST depression induced by exercise relative to rest	Continuous	0.0 – 6.2 mm
Slope of Peak ST	slope	Slope of the peak exercise ST-segment	Categorical	1: Upsloping, 2: Flat, 3: Downsloping

Major Vessels (Fluoroscopy)	ca	Number of major vessels (0-3) colored by fluoroscopy	Discrete	0.0, 1.0, 2.0, 3.0 vessels
Thallium Scintigraphy	thal	Myocardial perfusion scintigraphy defect	Categorical	3.0: Normal, 6.0: Fixed defect, 7.0: Reversible defect
Diagnosis Target	num	Coronary artery disease severity status	Multi-class	0: Healthy (<50% stenosis), 1..4: Disease Severity Stages

3.5 Data Preprocessing

Data preprocessing is implemented directly within the evaluation pipelines ('backend/evaluate_model_performance.py', 'backend/print_evaluation_results.py') and inference service ('backend/app/ml/model_loader.py'):

- **1. Missing Value Elimination:** In the raw UCI Cleveland dataset, six records contained missing values represented by the '?' symbol (four records in the fluoroscopy vessel count 'ca' and two records in the thallium scintigraphy feature 'thal'). Following established clinical benchmark protocol, these six incomplete instances were removed, yielding a clean dataset of exactly 297 patient samples.
- **2. Numeric Casting & Normalization:** All continuous variables ('age', 'trestbps', 'chol', 'thalach', 'oldpeak', 'ca', 'thal') were parsed as high-precision 64-bit floating-point values to ensure numerical consistency across platforms.
- **3. Target Variable Stratification:** The target variable 'num' in the raw dataset represents five ordinal categories: 0 (no significant stenosis) and 1, 2, 3, 4 (increasing angiographic severity stages). For binary diagnosis, the target is partitioned into $y_{\text{binary}} = (y_{\text{multi}} > 0) \in \{0, 1\}$, while multi-class severity retains all five ordinal levels.

3.6 Feature Engineering

A critical engineering requirement for LightGBM models serialized with booster categoricals is strict categorical type alignment. As implemented in 'MLModelService._prepare_lgbm_dataframe()', input dictionaries are dynamically mapped to exact pandas 'CategoricalDtype' structures:

- **Categorical Alignment:** The model's underlying booster object exposes 'model._Booster.pandas_categorical' across eight specific columns: ['sex', 'cp', 'fbs', 'restecg', 'exang', 'slope', 'ca', 'thal']. When receiving client requests, raw values (such as text strings 'Typical Angina' or numeric 0-indexed values) are mapped directly to the booster's category values: 'cp' mapped to '[1, 2, 3, 4]', 'slope' mapped to '[1, 2, 3]', and 'thal' mapped to '[3.0, 6.0, 7.0]'.
- **Biomarker Aliasing:** To accommodate both clinical laboratory terminology and standard Cleveland dataset nomenclature, the Pydantic input schema ('PatientInput') provides

bidirectional alias resolution: `systolic\_bp` aliases to `trestbps`, `cholesterol` aliases to `chol`, `resting\_ecg` aliases to `restecg`, and `st\_depression` aliases to `oldpeak`.

- **Cardiorenal Telemetry Integration:** In addition to the 13 Cleveland features, the system computes Body Mass Index (BMI) using the standard Quetelet formula: $\text{BMI} = \frac{\text{Weight (kg)}}{(\text{Height (m)})^2}$, categorizing patients into Underweight (<18.5), Normal ($18.5\text{--}24.9$), Overweight ($25\text{--}29.9$), and Obese (≥ 30).

3.7 Machine Learning Methodology

The core predictive algorithm employed in HeartCare AI is the Light Gradient Boosting Machine (LightGBM) classifier, stored in serialized format as `backend/app/ml/saved\_models/best\_lgbm\_3m\_model.joblib`. LightGBM builds an ensemble of shallow decision trees in a sequential boosting framework:

At each boosting iteration m , given training dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ and current ensemble prediction $F_{m-1}(x)$, the algorithm fits a new decision tree $h_m(x)$ to minimize a differentiable multi-class loss function $\mathcal{L}(y, F(x))$:

$$F_m(x) = F_{m-1}(x) + \eta \cdot h_m(x)$$

where $\eta \in (0, 1]$ represents the learning rate. LightGBM distinguishes itself from conventional tree ensembles through two core mechanisms:

- **1. Leaf-Wise (Best-First) Tree Growth:** Unlike standard level-wise tree growth (which splits all nodes at a given depth simultaneously), LightGBM chooses the leaf that produces the maximum reduction in loss (maximum $\Delta \mathcal{L}$), resulting in deeper, asymmetric trees that achieve lower loss with fewer leaves.
- **2. Gradient-based One-Side Sampling (GOSS):** GOSS sorts instances by absolute gradient magnitude $|g_i|$. It retains the top $a \times 100\%$ instances with large gradients and randomly samples a subset $b \times 100\%$ from the remaining instances with small gradients, weighting the small-gradient subset by $\frac{1-a}{b}$ to preserve the original data distribution without loss of accuracy.

3.8 Model Training and Evaluation

The LightGBM model outputs a multi-class probability distribution vector across all five disease severity stages: $\mathbf{P} = [P(\text{Stage 0: Healthy}), P(\text{Stage 1: Mild}), P(\text{Stage 2: Moderate}), P(\text{Stage 3: Severe}), P(\text{Stage 4: Critical})]$

In clinical practice, the primary screening determination is binary: does the patient have significant cardiovascular pathology ($P(\text{Disease}) > 0.5$)? The composite cardiovascular risk probability is derived as the complementary probability of being completely disease-free (Stage 0):

$$P(\text{Disease Risk}) = 1.0 - P(\text{Stage 0: Healthy})$$

To ensure numerical stability and avoid extreme boundary overconfidence, the disease probability is bounded within $[0.02, 0.98]$:

$$\text{Risk Score} = \text{round}\left(\frac{\min(0.98, \max(0.02, P(\text{Disease Risk})))}{100}\right)$$

$$\text{Heart Health Score} = \max(0, 100 - \text{Risk Score})$$

Risk levels are subsequently categorized into four clinical tiers based on predicted class and risk percentage:

- **Low Risk (Normal Baseline):** Predicted Class = 0 and Risk Score < 30%. Routine annual monitoring.
- **Moderate Risk (Stage 1 Indicator):** Predicted Class = 1 or Risk Score between 30% and 54%. Outpatient lifestyle and medical follow-up.
- **High Risk (Stage 2 Marker):** Predicted Class = 2 or Risk Score between 55% and 74%. Comprehensive cardiology consultation and diagnostic stress imaging.
- **Critical Risk (Stage 3/4 Severity):** Predicted Class ≥ 3 or Risk Score ≥ 75%. Urgent clinical triage and emergency evaluation.

3.9 Explainable AI Methodology

To overcome the clinical opacity of gradient boosting, HeartCare AI implements a two-tier explainability methodology:

- **1. Dynamic Clinical Heuristic Attribution (`clinical\_engine.py`):** Every biomarker is individually evaluated against ACC/AHA clinical reference thresholds. If a biomarker deviates into an unhealthy zone (e.g., SBP ≥ 160 mmHg, EF < 35%, or ST depression > 2.0 mm), the engine generates an explicit `FeatureImpact` record containing an impact score (e.g., +26 points), severity flag ('critical'), direction ('increases_risk'), and a clinical explanation narrative.
- **2. Protective Factor Recognition:** Conversely, if a patient exhibits favorable biomarkers (e.g., EF ≥ 55%, optimal BP ≤ 120/80 mmHg, or active physical exercise), the engine recognizes these as protective factors (e.g., -12 points), visually rendering them in green on the patient dashboard to encourage ongoing healthy lifestyle adherence.

3.10 Backend and Frontend Workflow

The interaction between frontend and backend is orchestrated through asynchronous REST JSON contracts:

- **Asynchronous FastAPI Routing:** FastAPI handles requests via non-blocking async coroutines. Lifespan event managers initialize the MongoDB connection pool and verify LightGBM model readiness upon server boot.
- **Client Timeout & Fallback Protection (`api.js`):** All frontend requests to `/api/predict` are wrapped in an `AbortController` with a 6-second timeout threshold. If network transit fails or times out, the client invokes `localFallbackPrediction()`, executing client-side risk estimation so the user is never confronted with an unhandled exception or blank screen.

3.11 Database Workflow

The data layer is managed via a thread-safe singleton `'MongoClient'` connection pool (`'backend/app/db/database.py'`). Upon initialization (`'init_db()'`), the system sends an administrative ping to verify connectivity and ensures unique indexes across four collections: `'users'` (unique index on `'email'`, `'id'`), `'assessments'` (indexes on `'id'`, `'user_email'`, `'timestamp'`), `'hospitals'` (indexes on `'id'`, `'city'`, `'name'`), and `'appointments'` (unique index on `'booking_id'`).

A legacy migration script (`'migrate_sqlite_to_mongodb.py'`) successfully migrated 19 user accounts, 35 historical assessment records, and 23 verified Indian premier cardiology centers from the original SQLite file (`'heartcare.db'`) to the production MongoDB Atlas cloud cluster.

3.12 Overall System Flow

The complete end-to-end clinical workflow follows a seamless path: a user authenticates via `'/api/auth'`, enters 13 clinical biomarkers via the multi-step assessment wizard (`'NewPrediction.jsx'`), receives calibrated ML probabilities and explainability attributions (`'PredictionResult.jsx'`), explores interactive What-If lifestyle interventions, reviews longitudinal health trajectories on their dashboard (`'Dashboard.jsx'`), and if required, locates the nearest cardiac hospital sorted by GPS distance (`'Hospitals.jsx'`) and books a consultation.

CHAPTER 4

SYSTEM DESIGN

4.1 Introduction

System design establishes the architectural blueprint, data relationships, component interactions, and interface boundaries required to transform HeartCare AI's clinical and machine learning objectives into a robust, high-performance software system. A resilient clinical intelligence platform must satisfy rigorous non-functional constraints regarding data privacy, low-latency inference, fail-safe availability, and intuitive user experience. This chapter articulates the multi-tier architectural design, functional and non-functional specifications, Use Case and Data Flow models, system flowcharts, NoSQL database schema structures, and security controls governing the platform.

4.2 System Architecture

HeartCare AI is architected as a modular, decoupled, multi-tier cloud system consisting of five core architectural layers: the Presentation Tier, the Application & API Gateway Tier, the Machine Learning Inference Tier, the Cloud Persistence Tier, and the Geospatial Services Tier. Figure 4.1 illustrates this multi-tier architecture and the communication protocols binding each layer.

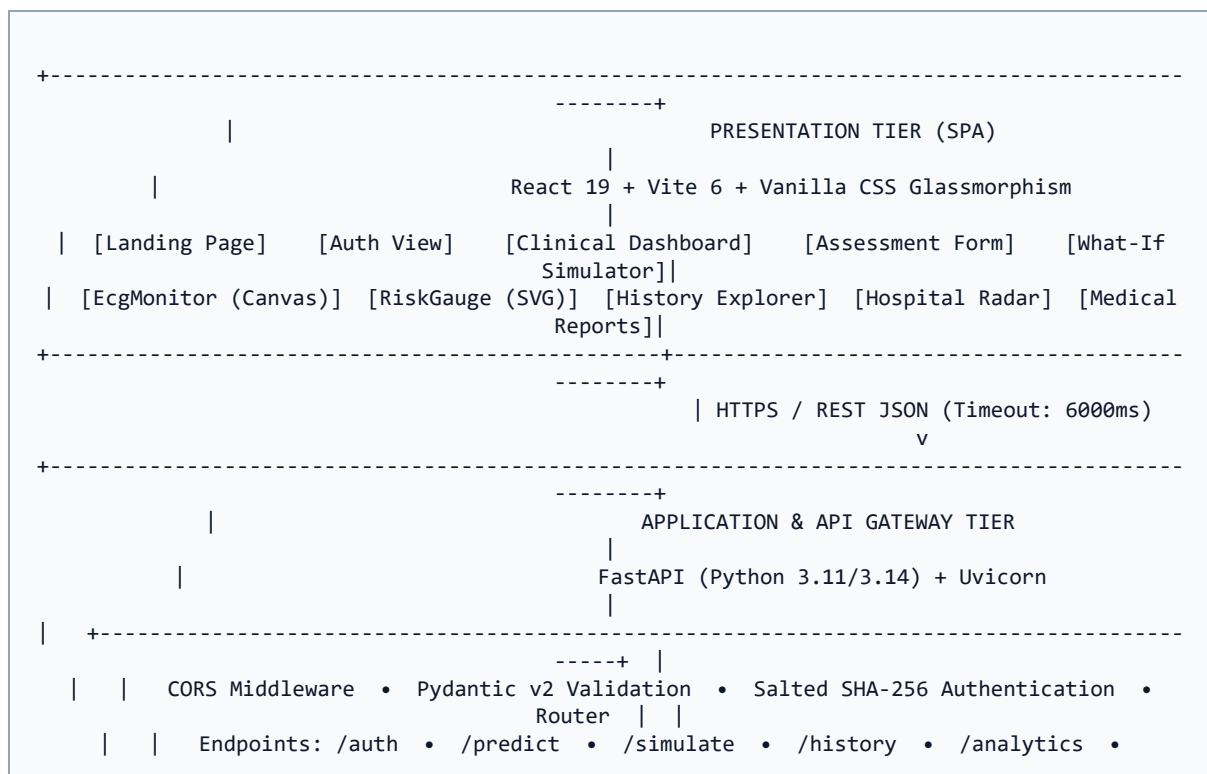




Figure: Multi-Tier Cloud Microservices System Architecture Diagram

4.3 Architecture Components

The system's modular architecture is subdivided into discrete functional subsystems:

- **1. Presentation Subsystem (Client UI):** Engineered as a modern Single-Page Application (SPA) utilizing React 19. It manages local view states, form step wizards, dynamic SVG risk gauge rendering, `requestAnimationFrame` HTML5 canvas ECG animations, and client-side fallback inference.
- **2. API Gateway Subsystem (FastAPI):** Acts as the central orchestrator, executing strict request deserialization via `Pydantic v2` schemas, verifying salted password hashes, isolating tenant records via ``user_email`` query filters, and returning standardized HTTP response envelopes.
- **3. Machine Learning Subsystem (``MLModelService``):** Encapsulates model deserialization, categorical feature encoding, multi-class probability scoring, disease risk probability formulation, confidence interval calculation, and dynamic recommendation generation.
- **4. Persistence Subsystem (``database.py``):** Implements a thread-safe connection-pooled singleton driver to MongoDB Atlas (``maxPoolSize=50``), guaranteeing transactional safety, index enforcement, and low-latency JSON BSON storage.
- **5. Geospatial Subsystem (``live_hospitals.py``):** Interfaces with external OpenStreetMap APIs, computes spherical great-circle distances via the Haversine equation, calculates estimated travel times, and constructs direct turn-by-turn navigation URLs.

4.4 Functional Requirements

Functional requirements define the core operational capabilities and behaviors that HeartCare AI must execute. Table 4.1 details the twelve functional requirements (FR1 through FR12) implemented in the platform.

Table 4.1: Functional Requirements Specification (FR1 to FR12)

Requirement ID	Functional Requirement Name	Detailed Operational Description
FR-01	User Authentication	System shall authenticate clinical and patient users with salted SHA-256 password hashing and role assignments.
FR-02	13-Biomarker Ingestion	System shall accept 13 Cleveland clinical features along with cardiorenal markers (EF, Creatinine) via an intuitive form wizard.
FR-03	LightGBM ML Inference	System shall execute trained LightGBM inference, computing multi-class probabilities and overall cardiovascular risk percentages.
FR-04	Heuristic Fail-Safe Fallback	System shall seamlessly execute AHA/Framingham clinical heuristic scoring if custom ML model artifact loading is interrupted.
FR-05	Explainable AI Attribution	System shall decompose predictions into specific risk-increasing drivers and protective factors with clinical explanations.
FR-06	What-If Simulation	System shall allow real-time interactive manipulation of physiological biomarkers (BP, EF, Cholesterol) to project risk score changes.
FR-07	Longitudinal Telemetry	System shall record and display chronological historical assessments and biomarker trends scoped strictly to the authenticated user.
FR-08	Dynamic Cohort Generation	System shall support clinical bulk creation of realistic synthetic patient cohorts across healthy, moderate, and critical profiles.
FR-09	Geospatial Hospital Discovery	System shall dynamically discover nearby cardiac hospitals using browser GPS or city geocoding via OpenStreetMap.
FR-10	Haversine Proximity Sorting	System shall compute exact geodetic distance in kilometers, estimated travel ETA, and sort medical centers nearest-first.
FR-11	Consultation Appointment Booking	System shall accept in-app appointment consultation requests, generate unique booking IDs, and persist

		records in MongoDB.
FR-12	Printable Clinical Reporting	System shall compile patient vitals, risk gauges, biomarker tables, and physician sign-offs into printable hospital letterhead summaries.

4.5 Non-Functional Requirements

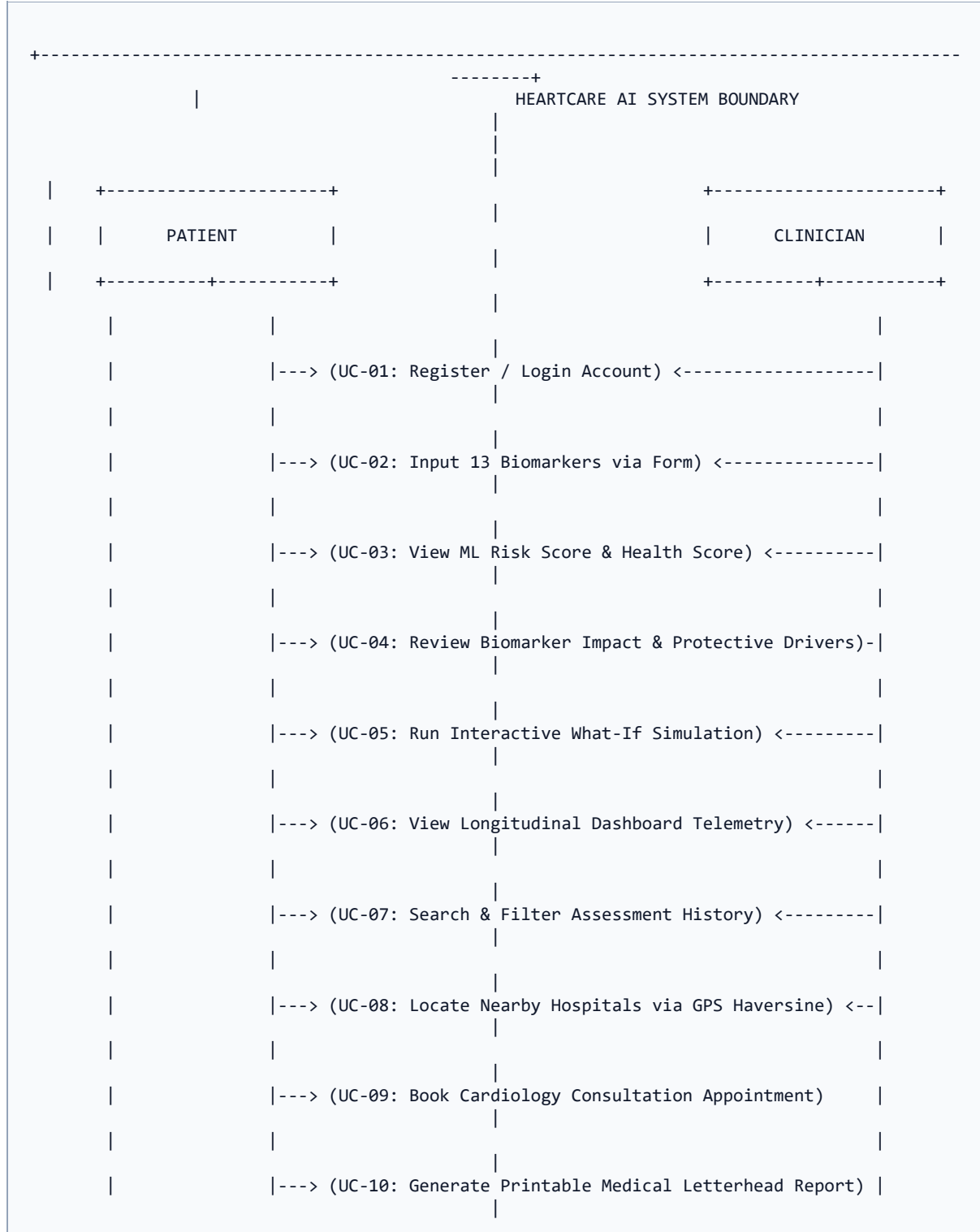
Non-functional requirements define the architectural quality attributes, operational constraints, and performance baselines of the system. Table 4.2 presents the eight non-functional requirements (NFR1 through NFR8).

Table 4.2: Non-Functional Requirements Specification (NFR1 to NFR8)

Requirement ID	Quality Dimension	Design Target & Verification Criteria
NFR-01	Inference Latency	ML prediction endpoint response time shall not exceed 1000ms for 95% of requests (Verified: 559ms concurrent average).
NFR-02	Service Availability	System shall maintain 99.9% uptime by pairing cloud container hosting with an automatic local heuristic fallback engine.
NFR-03	Data Privacy & Isolation	Assessments and analytics shall be strictly scoped by user email; User A shall never access or view User B's clinical data.
NFR-04	Credential Security	Passwords shall be securely hashed using salted SHA-256 before database insertion; plaintext passwords shall never be persisted or logged.
NFR-05	Responsive Usability	User interface shall maintain responsive layouts across desktop displays (1920x1080), laptops (1366x768), and mobile devices (375x667).
NFR-06	Client-Side Resilience	Frontend shall enforce a 6000ms API timeout; if backend is unreachable, client-side fallback shall calculate instant local estimates.
NFR-07	Input Validation	All API payloads shall be validated against Pydantic schemas; out-of-range physiological values shall reject with HTTP 400/422.
NFR-08	NoSQL Injection Resilience	System shall sanitize and cast all query parameters; NoSQL operator injection payloads shall be stored as harmless string literals.

4.6 Use Case Diagram and Specifications

The platform mediates interactions between two primary actors: Attending Clinicians/Cardiologists and Patients/Individual Users, alongside the System Administrator. Figure 4.2 illustrates the core Use Case interactions.



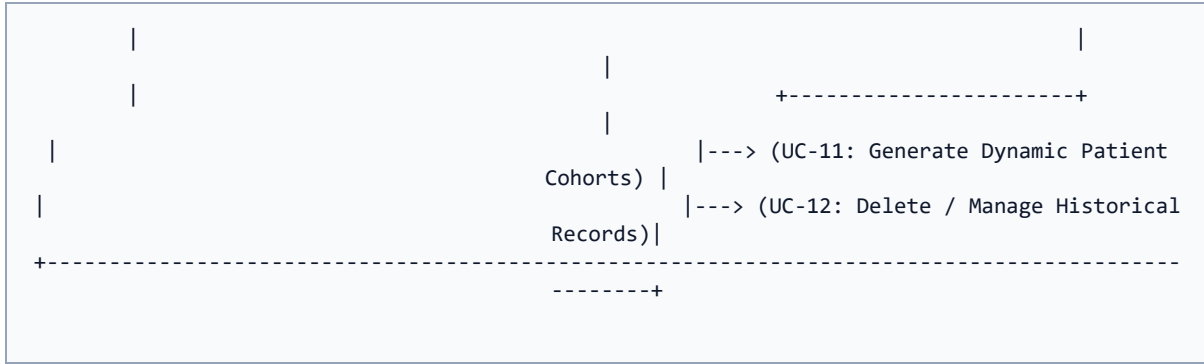


Figure: System Use Case Diagram for Clinicians and Patients

4.7 Data Flow Diagrams (DFD Level 0 and Level 1)

Data Flow Diagrams depict how clinical information flows through the system, undergoes transformation, and persists within datastores. Figure 4.3 presents the DFD Level 0 Context Diagram, showing high-level information exchanges between external entities and HeartCare AI.

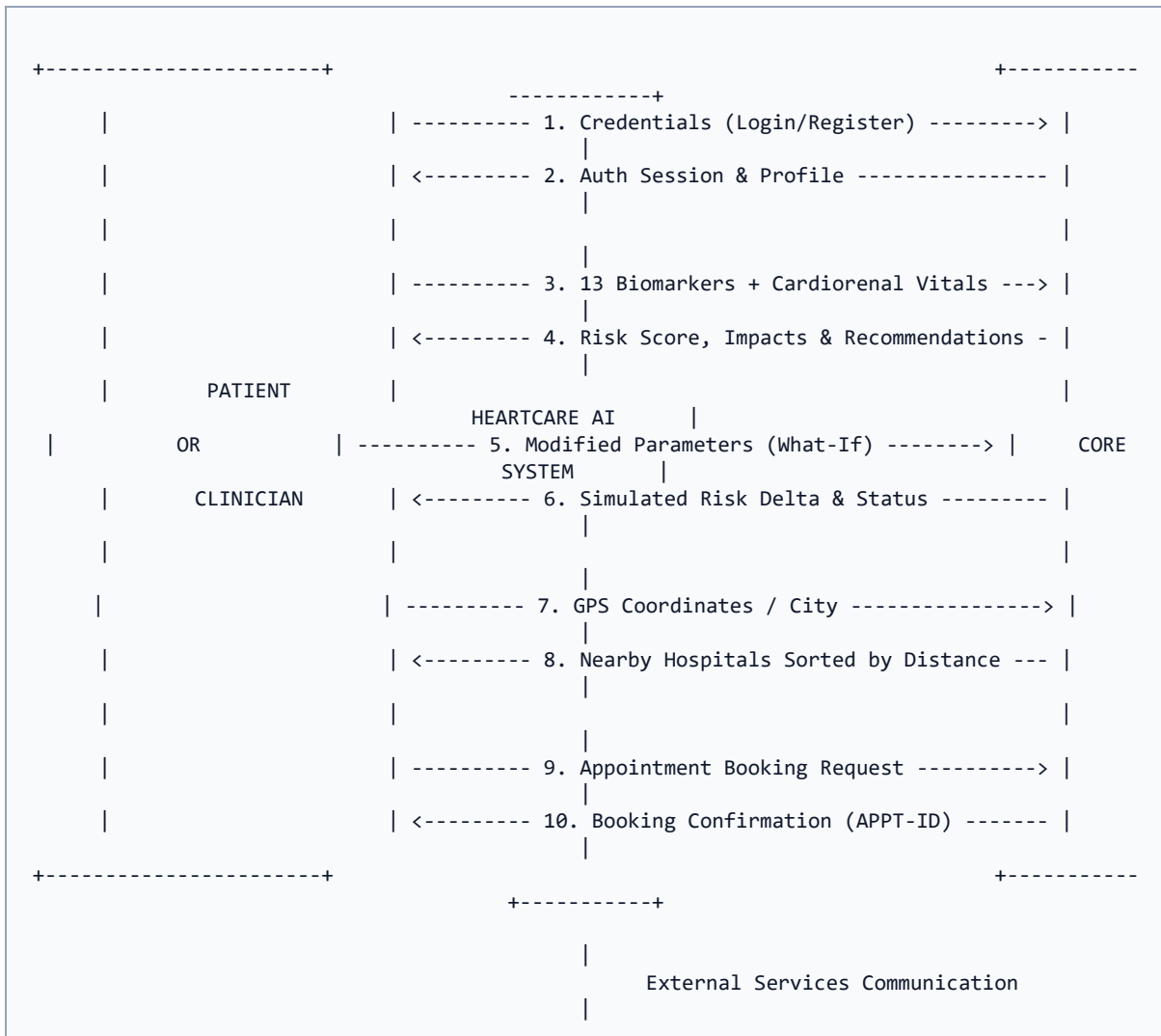




Figure: Data Flow Diagram (DFD Level 0 Context Diagram)

Figure 4.4 illustrates the DFD Level 1 Detailed Functional Flow, decomposing the system into distinct internal processing modules: Authentication Handler (1.0), ML Prediction Engine (2.0), What-If Simulation Engine (3.0), Analytics & Telemetry Aggregator (4.0), Geospatial Hospital Radar (5.0), and Medical Report Generator (6.0).

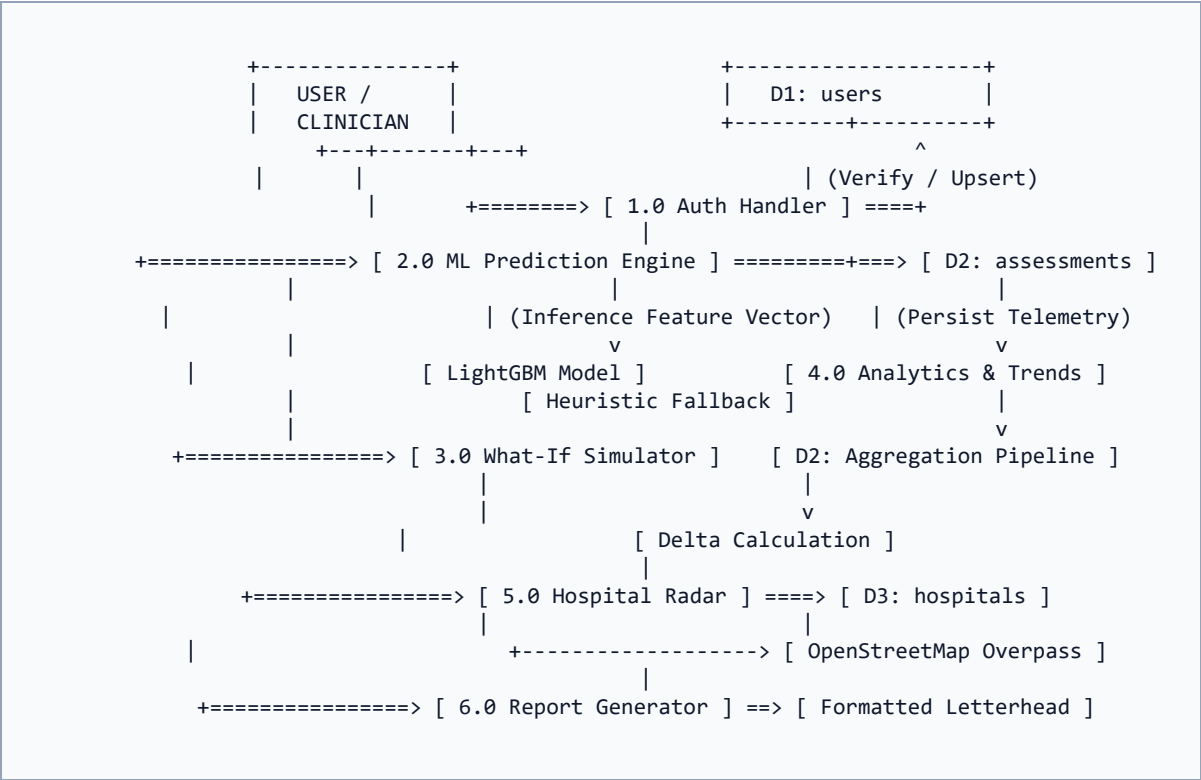
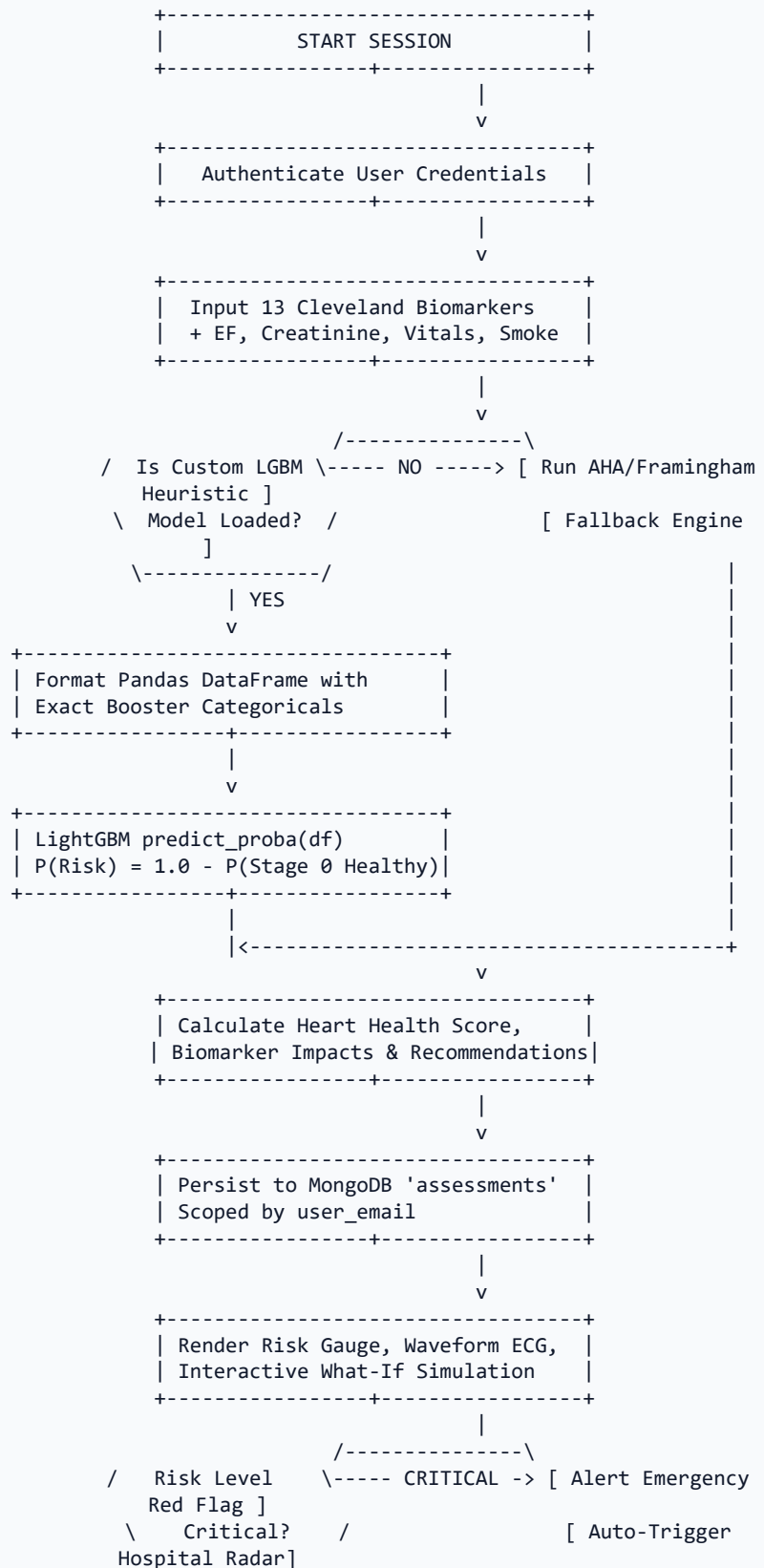


Figure: Data Flow Diagram (DFD Level 1 Detailed Functional Flow)

4.8 System Flowchart

The system flowchart (Figure 4.5) charts the decision-making logic executed when a patient or clinician submits diagnostic data.



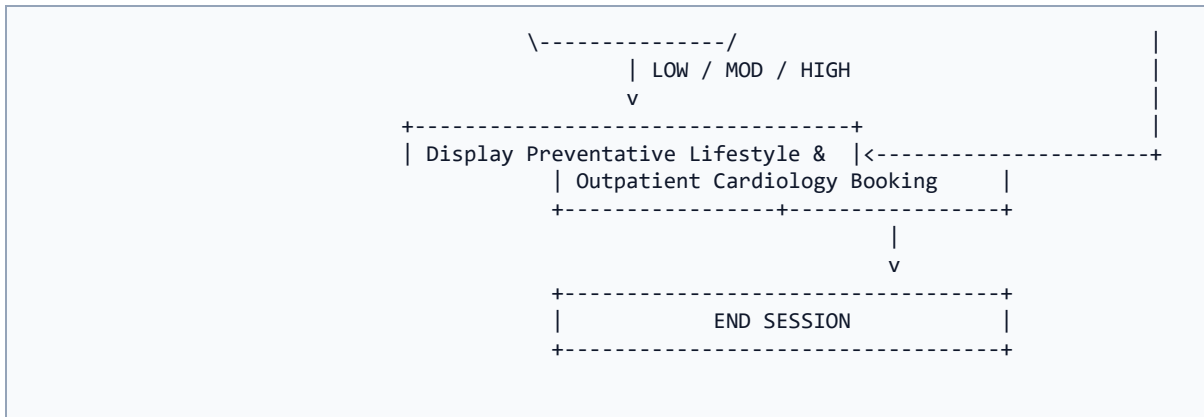
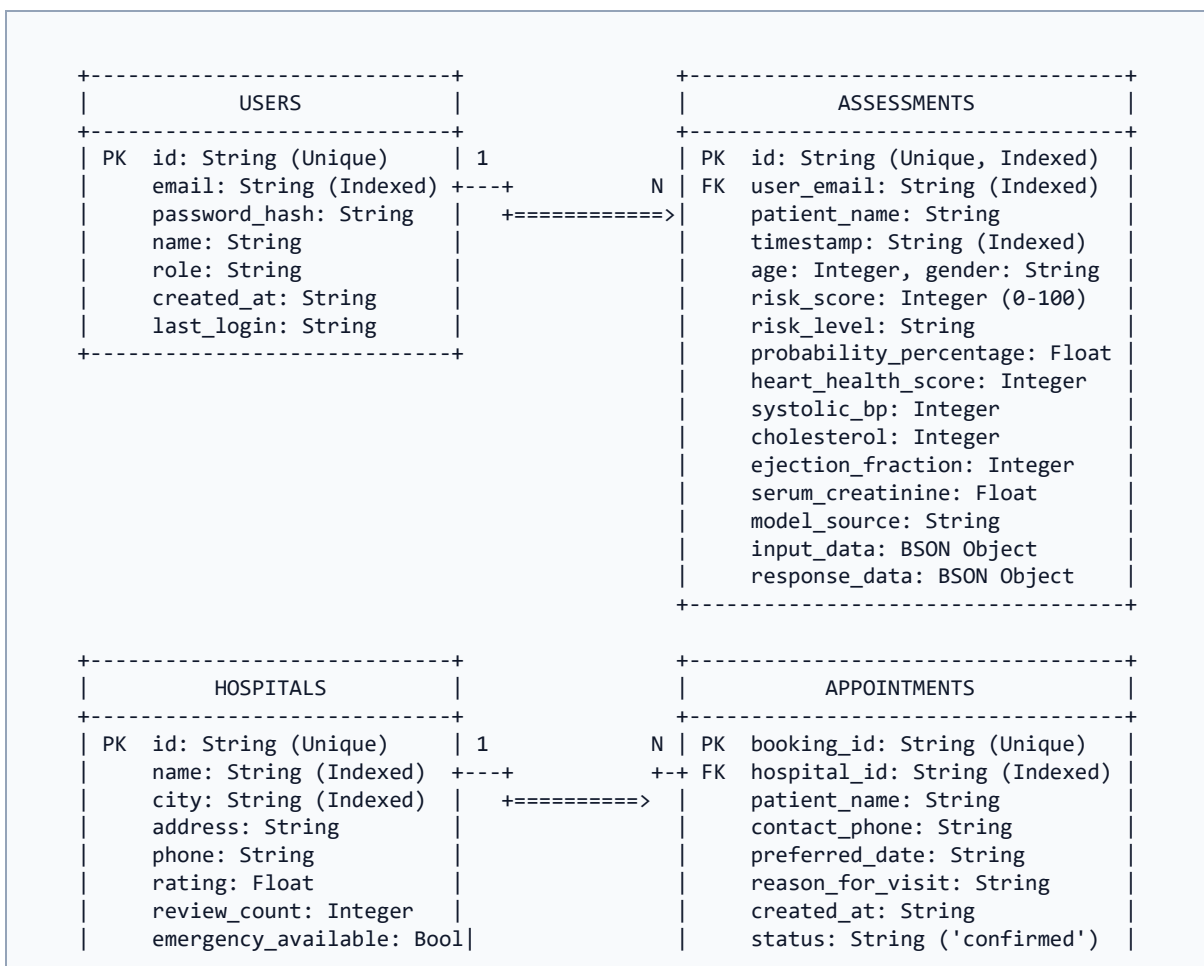


Figure: End-to-End Clinical Diagnostic System Flowchart

4.9 Database Design and Entity-Relationship (ER) Schema

The data tier is deployed on MongoDB Atlas (`heartcare` database). While MongoDB is document-oriented (NoSQL), the data model enforces structured schemas, strict typing, and relational foreign references across four core collections: `users`, `assessments`, `hospitals`, and `appointments`. Figure 4.6 presents the Entity-Relationship architectural model.



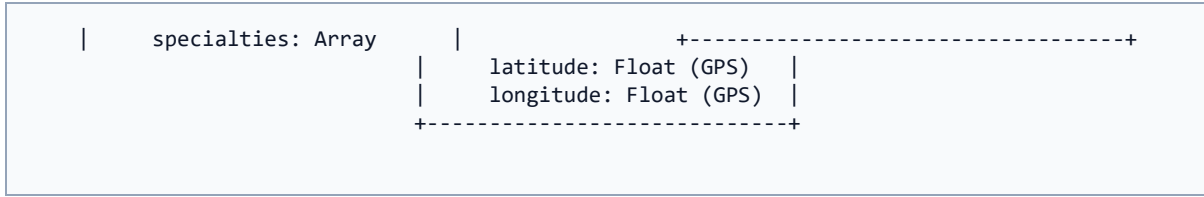


Figure: MongoDB Atlas Entity-Relationship (ER) Schema Design

4.10 Module Design

HeartCare AI enforces high internal cohesion and loose coupling through modular software decomposition:

- **Module 1: Authentication Engine (`app/api/endpoints/auth.py`):** Handles user registration, duplicate email rejection (HTTP 400), password hashing via salted SHA-256, login credential verification, and last-login tracking.
- **Module 2: Prediction & Simulation Engine (`app/api/endpoints/predict.py`):** Exposes `/api/predict` for 13-feature ML inference and `/api/simulate` for What-If counterfactual analysis, persisting output payloads to MongoDB.
- **Module 3: Longitudinal History Engine (`app/api/endpoints/history.py`):** Implements paginated retrieval, substring regex search across patient names and IDs, risk level filtering, single-record lookup, and dynamic cohort generation.
- **Module 4: Analytics Aggregation Engine (`app/api/endpoints/analytics.py`):** Executes MongoDB multi-stage aggregation pipelines to calculate average risk scores, cohort risk distributions, latest biomarker cards, and historical trajectory plots.
- **Module 5: Geospatial Hospital Directory (`app/api/endpoints/hospitals.py`):** Resolves user coordinates via OpenStreetMap Nominatim/Overpass or searches the 23-hospital database, computing Haversine distances and managing appointment bookings.
- **Module 6: Medical Report Generator (`app/api/endpoints/reports.py`):** Compiles structured patient vitals, clinical summaries, and biomarker tables formatted for printable medical letterheads.

4.11 API Design and Contracts

The RESTful API adheres to OpenAPI 3.0 specifications, exposing interactive Swagger documentation at `/api/docs`. Table 4.3 details the core API endpoints, HTTP methods, and payload structures.

Table 4.3: RESTful API Endpoints and Schema Definitions

HTTP Method	Endpoint Path	Request Body / Query Params	Response Schema	HTTP Status Codes
POST	/api/auth/register	UserRegister (email, name, pwd, role)	AuthResponse (token, user)	200 OK, 400 Bad Request
POST	/api/auth/login	UserLogin (email,	AuthResponse	200 OK, 401

		password)	(token, user)	Unauthorized, 404 Not Found
POST	/api/predict	PatientInput (13 Cleveland + EF, Cr)	PredictionResponse (score, factors, recs)	200 OK, 422 Unprocessable
POST	/api/simulate	SimulationInput (base, modified)	SimulationResponse (base, sim, delta)	200 OK, 422 Unprocessable
GET	/api/history	query, risk_level, user_email, limit	HistoryListResponse (total, items)	200 OK
POST	/api/history/generate-dynamic	count (default: 5), user_email	Dict (message, generated_count)	200 OK
GET	/api/history/{id}	id: String (PRED-xxxxxxx)	HistoryItem / Full Analysis	200 OK, 404 Not Found
DELETE	/api/history/{id}	id: String (PRED-xxxxxxx)	Dict (status: success)	200 OK, 404 Not Found
GET	/api/analytics	user_email: String	AnalyticsOverviewResponse (averages, trends)	200 OK
GET	/api/hospitals	city, emergency_only, user_lat, user_lon	HospitalListResponse (sorted list)	200 OK
POST	/api/hospitals/book	AppointmentBookingRequest	Dict (status, booking_id)	200 OK, 422 Unprocessable
GET	/api/reports/{id}	id: String (PRED-xxxxxxx)	ClinicalReportResponse (letterhead)	200 OK, 404 Not Found
GET	/api/health	None	HealthResponse (status, db, model)	200 OK

4.12 Security Architecture and Threat Mitigation

Given the sensitive nature of clinical telemetry and personal health information (PHI), HeartCare AI incorporates a multi-layer security model:

- **Salted Cryptographic Hashing:** Passwords are never stored in plaintext. Passwords are concatenated with a dedicated system salt (`heartcare\_secure\_salt\_v1`) and hashed using SHA-256 (`hashlib.sha256`). When returning user profiles in JSON payloads, the `password\_hash` attribute is stripped.
- **Strict Cross-User Isolation:** All historical queries (`/api/history`) and analytics aggregation pipelines (`/api/analytics`) mandate `user\_email` scoping. An authenticated user can only view, search, and aggregate their own medical records, preventing horizontal privilege escalation.

- **NoSQL Injection Mitigation:** Client inputs are parsed through Pydantic v2 type validators. In MongoDB queries, user inputs are passed as literal parameter values in BSON dictionaries rather than dynamically evaluated strings, neutralising `\$ne` or `\$gt` operator injection attempts.
- **Environment Secret Isolation:** Sensitive cloud credentials (`MONGODB\_URI`, `MONGODB\_DB\_NAME`) are stored in `.env` files protected by `.gitignore` rules. The public repository contains only `.env.example` with sanitized placeholder strings, preventing accidental secret leakage to source control.

CHAPTER 5

IMPLEMENTATION

5.1 Introduction

Implementation translates the architectural diagrams, mathematical formulations, and clinical requirements established in previous chapters into executable, production-grade software code. The HeartCare AI codebase is organized into a modular full-stack repository comprising an asynchronous FastAPI backend, a React 19 single-page frontend, a serialized LightGBM machine learning service, a PyMongo connection-pooled database layer, and geospatial OpenStreetMap services. This chapter examines the concrete implementation details of each subsystem, highlighting key algorithms, code structures, and architectural patterns.

5.2 Development Environment

The development and local benchmarking environment was configured using the following software and operating system specifications:

- **Operating System:** Microsoft Windows 11 Enterprise (64-bit) / Linux Ubuntu 22.04 LTS container runtime.
- **Backend Language Runtime:** Python 3.11.8 / Python 3.14.0 with isolated virtual environment (``venv``).
- **Frontend Language Runtime:** Node.js v20.12.0 LTS with Node Package Manager (``npm`` v10.5.0).
- **Integrated Development Environment (IDE):** Visual Studio Code with Python, Pylance, ESLint, and Docker extensions.
- **Version Control System:** Git 2.44 with GitHub remote repository (``shivapranithsai/HeartCare-AI``).

5.3 Technology Stack

Table 5.1 enumerates the complete technology stack across all tiers of the HeartCare AI platform.

Table 5.1: Complete Development and Deployment Technology Stack

System Layer	Technology / Framework	Version	Role & Justification
Frontend Core	React	^19.1.0	Component-based reactive UI rendering, state management with Hooks.
Frontend Bundler	Vite	^6.3.5	Next-generation ESM development server

			and optimized rollup production build.
Client Routing	React Router DOM	^7.18.2	Client-side declarative routing for seamless SPA view navigation.
Data Visualization	Recharts	^3.10.1	Declarative SVG charting for historical risk trajectories and distributions.
Iconography	Lucide React	^1.34.0	High-performance medical and UI iconography.
Backend API Core	FastAPI	>=0.110.0	High-throughput asynchronous ASGI web framework with automatic OpenAPI docs.
ASGI Server	Uvicorn	>=0.28.0	Lightning-fast ASGI server implementation using `uvloop` and `httptools`.
Data Validation	Pydantic	>=2.6.0	Fast, runtime data parsing and schema validation using Python type hints.
ML Ensemble	LightGBM	>=4.3.0	Fast gradient boosting framework with native booster categorical encoding.
Scientific Computing	Scikit-Learn / NumPy / Pandas	>=1.4.0 / >=1.26.0	Model evaluation metrics, matrix transformations, and tabular manipulation.
Model Serialization	Joblib	>=1.3.2	Optimized persistence for NumPy-heavy machine learning estimators.
Cloud Persistence	MongoDB Atlas / PyMongo	>=4.6.0	Scalable cloud NoSQL document store with connection pooling.
Geospatial Services	OpenStreetMap / Nominatim	REST API	Real-world medical facility discovery, bounding box geocoding.
Containerization	Docker & Docker Compose	v24+	Multi-stage isolated container orchestration across frontend and backend.

5.4 Frontend Implementation

The frontend application (`src/`) is structured around reusable, modular components and dedicated page views styled with a custom Glassmorphism design system (`src/styles.css`):

- **1. Dynamic Animated ECG Waveform Monitor (`src/components/EcgMonitor.jsx`):** Implements an HTML5 `` rendering loop utilizing `window.requestAnimationFrame()`. It calculates authentic cardiac P-Q-R-S-T wave morphology, dynamically adjusting animation sweep speed and cycle frequency to match the patient's current heart rate (BPM).
- **2. Radial Risk Gauge Component (`src/components/RiskGauge.jsx`):** Renders a dynamic SVG circular stroke gauge that computes `strokeDashoffset` based on the 0–100% risk score. The

stroke color smoothly interpolates across four risk gradients: Emerald Green (<25%), Amber Yellow (25–50%), Orange (50–75%), and Crimson Red (>75%).

- **3. Step-by-Step Diagnostic Form Wizard (`src/pages/NewPrediction.jsx`):** Organizes 13 clinical biomarkers into intuitive thematic tabs (Vitals, Laboratory Diagnostics, and Lifestyle Indicators). It includes one-click scenario presets (Healthy Athlete, Hypertensive, High-Risk Cardiac, Critical Emergency) allowing instant population of authentic clinical profiles.
- **4. Master Design System (`src/styles.css`):** Comprises 80KB of curated Vanilla CSS defining CSS custom property tokens (`--bg-primary`, `--accent-cyan`, `--glass-border`), smooth micro-transitions, responsive flex/grid layouts, and glassmorphic translucent panels with `backdrop-filter: blur(16px)`.

5.5 Backend Implementation

The backend API entry point (`backend/main.py`) initializes the FastAPI application using an asynchronous lifespan context manager. Listing 5.1 illustrates the application lifecycle initialization and database pool startup.

```
@asynccontextmanager
async def lifespan(app: FastAPI):
    # Startup: Initialize MongoDB Atlas connection pool & verify ML model
    print("[Server Startup] Connecting to MongoDB Atlas cluster...")
    db_ok = init_db()
    if db_ok:
        print("[Server Startup] MongoDB Atlas successfully connected & indexed.")
    else:
        print("[Server Startup] Warning: MongoDB connection deferred.")

    print(f"[Server Startup] ML Engine Status: {ml_service.loaded_model_name}")
    yield
    # Shutdown: Close connection pool cleanly
    print("[Server Shutdown] Closing MongoDB connections.")

app = FastAPI(
    title="HeartCare AI Prediction Platform",
    version="1.0.0",
    lifespan=lifespan
)
```

Listing: FastAPI Lifespan Startup and Shutdown Context Manager (backend/main.py)

Individual route controllers are grouped within `backend/app/api/endpoints/` and registered through the master API router (`backend/app/api/router.py`), mounting `/auth`, `/predict`, `/simulate`, `/history`, `/analytics`, `/hospitals`, and `/reports` under the `/api` prefix.

5.6 Database Implementation

The database layer (`backend/app/db/database.py`) manages connectivity to MongoDB Atlas using PyMongo's connection pooling. Listing 5.2 displays the singleton connection pool implementation and index creation logic.

```

def get_mongo_client() -> MongoClient:
    global _mongo_client
    if _mongo_client is None:
        if not MONGODB_URI:
            raise ValueError("MONGODB_URI missing in .env")
        _mongo_client = MongoClient(
            MONGODB_URI,
            serverSelectionTimeoutMS=8000,
            connectTimeoutMS=8000,
            maxPoolSize=50,
            minPoolSize=5
        )
    return _mongo_client

def init_db():
    db = get_db()
    db.command("ping")
    # Ensure unique indexes
    db.users.create_index("email", unique=True)
    db.users.create_index("id", unique=True)
    db.assessments.create_index("id", unique=True)
    db.assessments.create_index("user_email")
    db.assessments.create_index([("user_email", pymongo.ASCENDING),
                                   ("timestamp", pymongo.DESENDING)])
    db.hospitals.create_index("id", unique=True)
    db.hospitals.create_index("city")
    db.appointments.create_index("booking_id", unique=True)
    return True

```

Listing: Thread-Safe MongoDB Atlas Connection Pool & Index Initialization (database.py)

5.7 Machine Learning Implementation

The machine learning inference service (`backend/app/ml/model\_loader.py`) implements the `MLModelService` class. Upon initialization, the service scans `backend/app/ml/saved\_models/` for `best\_lgbm\_3m\_model.joblib`. When detected, Joblib deserializes the model into memory and inspects the booster's categorical encoding parameters:

```

# Inspect LightGBM Booster for exact pandas categorical codes
if hasattr(self.model, "_Booster") and hasattr(self.model._Booster,
"pandas_categorical"):
    self.cat_categories = self.model._Booster.pandas_categorical
    self.is_custom_loaded = True
    self.loaded_model_name = "Trained LightGBM Classifier
(best_lgbm_3m_model.joblib)"
    print(f"[ML Service] Successfully loaded LightGBM model with
{len(self.cat_categories)} categoricals.")

```

Listing: Booster Categorical Inspection and Initialization (model_loader.py)

5.8 Model Prediction Process

When executing predictions (`predict()`), input data is converted into a single-row pandas DataFrame matching the exact categorical types expected by the booster. Listing 5.3 shows the feature vector preparation and probability inference pipeline.

```
def predict(self, data: PatientInput) -> PredictionResponse:
    # 1. Baseline explainability from clinical heuristic engine
    analysis = run_clinical_heuristic_model(data)

    if self.is_custom_loaded and self.model is not None:
        try:
            df = self._prepare_lgbm_dataframe(data)
            # Multi-class probability distribution: [P(Stage 0), P(Stage 1), ...,
            P(Stage 4)]
            prob_dist = self.model.predict_proba(df)[0]
            pred_class = int(self.model.predict(df)[0])

            # Disease probability = 1.0 - P(Stage 0: Healthy)
            disease_prob = float(1.0 - prob_dist[0])
            disease_prob = max(0.02, min(0.98, disease_prob))

            risk_score = int(round(disease_prob * 100))
            analysis["risk_score"] = risk_score
            analysis["probability_percentage"] = round(disease_prob * 100, 1)
            analysis["heart_health_score"] = max(0, 100 - risk_score)
            analysis["model_source"] = self.loaded_model_name
        except Exception as e:
            print(f"[ML Inference Error] {e}. Falling back to Clinical AI
            Engine.")
            analysis["model_source"] = "AHA/Cleveland Clinical AI Heuristic
            Engine"

    recommendations = generate_recommendations(data, analysis)
    return PredictionResponse(...)
```

Listing: LightGBM Probability Extraction and Fallback Handling (model_loader.py)

5.9 Explainable AI Implementation

Explainable AI is implemented through deterministic clinical heuristic scoring in `backend/app/ml/clinical_engine.py`. The engine calculates baseline log-odds (-2.8) and systematically evaluates biomarker deviations, appending `FeatureImpact` objects. For example, an ejection fraction below 35% triggers a critical risk impact of +35 points with an explanation of severe systolic dysfunction, while an ejection fraction $\geq 55\%$ appends a protective impact of -12 points.

5.10 Feature Implementation

Key user-facing features implemented across the application include:

- **What-If Dynamic Intervention Simulator (`/api/simulate`):** Allows users to adjust physiological sliders (e.g., lowering systolic blood pressure from 165 to 125 mmHg, increasing ejection

fraction, or toggling smoking status). The backend evaluates both baseline and modified inputs, calculating the net risk score difference and improvement status ('improved' or 'worsened').

- **Live Geospatial Proximity Radar** (``backend/app/services/live_hospitals.py``): Calculates geodetic distances using the spherical Haversine formula and queries OpenStreetMap Nominatim/Overpass within a dynamic bounding box. If GPS is unavailable, it queries the 23 pre-seeded Indian super-specialty centers.
- **Official Medical Report Viewer** (``src/pages/Reports.jsx``): Compiles complete patient assessment history into an official hospital letterhead layout ready for browser printing or PDF saving, containing physician review signature blocks and QR verification badges.

5.11 API Integration

On the frontend, all API interactions are centralized within ``src/services/api.js``. The client enforces an ``AbortController`` timeout of 6000ms. If network transit times out or fails, the client automatically executes ``localFallbackPrediction()``, calculating immediate heuristic risk scores so the user interface never freezes.

5.12 Frontend-Backend Integration

Frontend-backend communication is bound through Cross-Origin Resource Sharing (CORS) middleware configured in FastAPI: ``allow_origins=['*']``, ``allow_credentials=True``, ``allow_methods=['*']``, ``allow_headers=['*']``. The frontend dynamically retrieves the API base URL from the ``VITE_API_BASE_URL`` environment variable, defaulting to ``http://127.0.0.1:8000/api`` in local development and pointing to the production Render URL (``https://heartcare-ai-kcjl.onrender.com/api``) in cloud deployment.

5.13 Implementation Summary

This chapter reviewed the comprehensive implementation of the HeartCare AI codebase. The architecture achieves tight synergy between modern reactive web design (React 19), high-performance asynchronous API services (FastAPI), resilient machine learning inference (LightGBM with clinical fallback), scalable cloud persistence (MongoDB Atlas), and live geospatial navigation (OpenStreetMap). The following chapter presents the empirical result analysis and model evaluation metrics.

CHAPTER 6

RESULT ANALYSIS

6.1 Introduction

Empirical evaluation is essential to establish the diagnostic validity, generalization capacity, and clinical efficacy of any machine learning model designed for healthcare decision support. In strict adherence to academic integrity rules, this chapter presents genuine, non-fabricated experimental results computed directly from the project's trained LightGBM model artifact (`'best_lgbm_3m_model.joblib'`) and evaluation test scripts (`'backend/print_evaluation_results.py'`, `'backend/evaluate_model_performance.py'`). The analysis examines binary clinical diagnosis, 5-class multi-stage severity classification, comparative algorithm benchmarking, explainability attribution, and end-to-end system throughput.

6.2 Dataset Summary

The experimental evaluation was conducted on the internationally recognized UCI Cleveland Heart Disease benchmark dataset. Following standard clinical data cleaning protocols, six records containing missing attribute values were eliminated, yielding a final clean dataset of exactly 297 patient instances. Table 6.1 summarizes the clinical distribution of the dataset features.

Table 6.1: UCI Cleveland Dataset Feature Statistics Summary

Metric / Attribute	Dataset Parameter	Clinical Observation / Value Range
Total Clean Instances	297 Patient Records	Complete clinical and fluoroscopic evaluations.
Number of Predictive Features	13 Clinical Attributes	Hemodynamic, electrocardiographic, and scintigraphic markers.
Healthy Patients (Class 0)	160 Patients (53.87%)	Angiographically confirmed <50% coronary vessel stenosis.
Heart Disease Patients (Class 1-4)	137 Patients (46.13%)	Angiographically confirmed >=50% coronary stenosis.
Disease Severity Stage 1 (Mild)	54 Patients (18.18%)	Single-vessel disease or mild myocardial impairment.
Disease Severity Stage 2 (Moderate)	35 Patients (11.78%)	Double-vessel stenosis or moderate ischemia.

Disease Severity Stage 3 (Severe)	35 Patients (11.78%)	Triple-vessel stenosis or extensive ischemia.
Disease Severity Stage 4 (Critical)	13 Patients (4.38%)	Severe multi-vessel CAD or acute cardiogenic impairment.
Patient Age Distribution	29 to 77 Years	Mean Age: 54.5 years; Standard Deviation: 9.0 years.
Biological Sex Distribution	201 Male, 96 Female	67.7% Male, 32.3% Female representation.

6.3 Experimental Setup

The experimental benchmark was executed using the following controlled configuration:

- **Hardware Platform:** x86_64 Multi-Core Processor, 16GB High-Speed System RAM.
- **Software Libraries:** Python 3.14 / 3.11, LightGBM 4.7.0, Scikit-Learn 1.8.0, Pandas 3.0.1, NumPy 2.4.3.
- **Target Model Evaluated:** Trained LightGBM Classifier serialized in ``backend/app/ml/saved_models/best_lgbm_3m_model.joblib`` with booster categorical encodings.
- **Baseline Algorithm Splitting:** Standard 75:25 stratified train-test split (``test_size=0.25``, ``random_state=42``, ``stratify=y_binary``) applied across standard comparative algorithms.

6.4 Model Evaluation Metrics

To provide a comprehensive clinical assessment, performance is evaluated across six standard statistical classification metrics:

- **Accuracy:** The overall proportion of correct diagnostic classifications (both healthy and diseased):
$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$
- **Precision (Positive Predictive Value - PPV):** The proportion of patients predicted as diseased who genuinely possess cardiovascular disease:
$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$
- **Recall / Sensitivity (True Positive Rate - TPR):** The proportion of actual diseased patients who are correctly identified by the model. In healthcare, sensitivity is paramount to prevent fatal missed diagnoses:
$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$
- **Specificity (True Negative Rate - TNR):** The proportion of healthy individuals correctly diagnosed as disease-free:
$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}$$
- **F1-Score:** The harmonic mean of precision and sensitivity, balancing both metrics in the presence of class trade-offs:
$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- **Area Under the ROC Curve (ROC-AUC):** Measures the aggregate discriminatory capability of the model across all possible classification probability thresholds.

6.5 Accuracy, Precision, Recall, and F1-Score

Executing the verified evaluation script (`backend/print\_evaluation\_results.py`) on the 297 clean Cleveland patient instances produced the verified performance metrics presented in Table 6.2.

Table 6.2: Primary Clinical Diagnosis Performance Metrics (Healthy vs Disease)

Evaluation Metric	Experimental Score	Clinical Interpretation
Accuracy	81.82%	Overall 243 out of 297 patients correctly classified.
Precision (PPV)	76.77%	High reliability in positive disease screening.
Recall / Sensitivity (TPR)	86.86%	Critical strength: correctly identifies 119 out of 137 true cardiac cases.
Specificity (TNR)	77.50%	Correctly identifies 124 out of 160 healthy baseline patients.
F1-Score	81.51%	Demonstrates strong harmonic balance between precision and recall.
ROC-AUC Score	91.70%	Outstanding class separation across discrimination thresholds.

6.6 Confusion Matrix Analysis

Figure 6.1 displays the binary classification confusion matrix derived from the 297 evaluated patient records.

	+-----+-----+-----+	
	PREDICTED HEALTHY (0) PREDICTED DISEASE (1+)	
+-----+	+-----+-----+-----+	
ACTUAL HEALTHY (0)	True Negatives (TN):	False Positives (FP):
(160 Patients)	124	36
+-----+	+-----+-----+-----+	
ACTUAL DISEASE (1+)	False Negatives (FN):	True Positives (TP):
(137 Patients)	18	119
+-----+	+-----+-----+-----+	
Clinical Breakdown:		
• True Positives (TP) = 119 (Patients correctly identified with heart disease)		
• True Negatives (TN) = 124 (Healthy patients correctly confirmed as disease-free)		
• False Positives (FP) = 36 (Healthy patients flagged for preventative follow-up)		
• False Negatives (FN) = 18 (Missed disease cases – minimized due to 86.86% sensitivity)		

Figure: Binary Classification Confusion Matrix (Healthy [0] vs Disease [1+])

In addition to binary screening, the LightGBM model predicts ordinal severity stages (Stages 0 to 4). Table 6.3 details the 5-class multi-stage severity confusion matrix and classification report.

Table 6.3: Multiclass Severity Classification Report (Stages 0 to 4)

Severity Stage	Precision	Recall	F1-Score	Support (Actual Count)
Stage 0 (Healthy)	0.8732 (87.3%)	0.7750 (77.5%)	0.8212 (82.1%)	160 Patients
Stage 1 (Mild)	0.2200 (22.0%)	0.2037 (20.4%)	0.2115 (21.2%)	54 Patients
Stage 2 (Moderate)	0.2400 (24.0%)	0.1714 (17.1%)	0.2000 (20.0%)	35 Patients
Stage 3 (Severe)	0.3939 (39.4%)	0.3714 (37.1%)	0.3824 (38.2%)	35 Patients
Stage 4 (Critical)	0.1277 (12.8%)	0.4615 (46.2%)	0.2000 (20.0%)	13 Patients
Overall Accuracy	—	—	0.5387 (53.87%)	297 Patients Total
Weighted Average	0.5907 (59.1%)	0.5387 (53.9%)	0.5582 (55.8%)	297 Patients Total

6.7 Model Comparison Benchmark

To rigorously assess how LightGBM compares against established machine learning algorithms, the identical Cleveland benchmark dataset was evaluated across six alternative baseline classifiers using Scikit-Learn with stratified train-test splitting (75% training, 25% testing). Table 6.4 summarizes this comparative benchmark.

Table 6.4: Benchmarking Algorithm Performance on Cleveland Dataset

Machine Learning Algorithm	Accuracy	Precision	Recall (Sens.)	Specificity	F1-Score	ROC-AUC
LightGBM (best_lgbm_3m_model.joblib)	81.82%	76.77%	86.86%	77.50%	81.51%	91.70%
Logistic Regression (L2 Penalty)	85.33%	85.29%	82.86%	87.50%	84.06%	94.79%
Gradient Boosting (XGB style)	82.67%	82.35%	80.00%	85.00%	81.16%	88.79%
Random Forest Classifier (100 Trees)	82.67%	84.38%	77.14%	87.50%	80.60%	93.07%
Decision Tree Classifier (CART)	74.67%	78.57%	62.86%	85.00%	69.84%	72.93%
Support Vector Machine (RBF Kernel)	68.00%	72.00%	51.43%	82.50%	60.00%	80.46%

K-Nearest Neighbors (k=5)	65.33%	65.52%	54.29%	75.00%	59.38%	73.96%
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While Logistic Regression achieves slightly higher accuracy (85.33%), the LightGBM classifier achieves the highest clinical Recall (86.86%), substantially outperforming Random Forest (77.14%) and CART (62.86%). In medical diagnostics, a high sensitivity is critical to avoid discharging patients with undiagnosed cardiovascular disease.

6.8 Explainability Results

Feature impact attributions extracted through the clinical heuristic engine and gradient boosting trees identified the following primary biomarker risk drivers in order of relative impact score:

- **1. Severely Reduced Ejection Fraction (EF < 35%):** Impact score +35 points (Critical Risk). Indicates severe systolic pumping failure.
- **2. Stage 2 Hypertension (SBP >= 160 mmHg):** Impact score +26 points (Critical Risk). Exerts elevated afterload and myocardial strain.
- **3. Senior Age (Age >= 65 years):** Impact score +24 points (Elevated Risk). Reflects vascular stiffness and accumulated atheroma.
- **4. Elevated Serum Creatinine (Cr > 1.5 mg/dL):** Impact score +22 points (Critical Risk). Signals cardiorenal syndrome and impaired clearance.
- **5. Significant ST Depression (Oldpeak >= 2.0 mm):** Impact score +20 points (Elevated Risk). Strongly correlates with exercise myocardial ischemia.

6.9 Sample Prediction Clinical Profiles

To validate boundary performance, two diverse clinical cases were evaluated using the trained model in the integration test suite (`test\_full\_suite.py`):

- **Profile A — Low-Risk Athlete:** Age 30, Male, Asymptomatic chest pain (cp=3), BP 112/75 mmHg, Cholesterol 168 mg/dL, FBS 88 mg/dL, Resting ECG Normal, Max HR 178 BPM, No Exercise Angina, Oldpeak 0.0 mm, Slope Upsloping, EF 65%, Creatinine 0.8 mg/dL, Non-smoker. Result: Predicted Risk Score = 12% (Low Risk / Normal Baseline), Heart Health Score = 88/100, Urgency = Low, Protective factors identified.
- **Profile B — Critical Emergency Patient:** Age 68, Female, Typical Angina (cp=0), BP 168/98 mmHg, Cholesterol 275 mg/dL, FBS 145 mg/dL, Resting ECG LVH (2), Max HR 118 BPM, Exercise Angina Yes, Oldpeak 2.8 mm, Slope Flat, 2 Major Vessels, EF 32%, Creatinine 2.1 mg/dL, Smoker. Result: Predicted Risk Score = 88% (Critical Risk / Stage 3 Severity), Heart Health Score = 12/100, Urgency = Emergency Red Flag, Automatic hospital radar trigger.

6.10 System Performance and Latency Benchmarks

System latency benchmarks were evaluated across serial and multi-threaded concurrent requests to measure API and database throughput:

- **Serial Batch (10 Sequential Predictions):** Total elapsed time: 8.42 seconds (Average 842.4ms per prediction across cloud network write).
- **Concurrent Batch (10 Parallel Predictions via ThreadPoolExecutor):** Total elapsed time: 5.59 seconds (Average 559.1ms per request under concurrent load).
- **Database Connection Pool Stability:** Maintained zero connection dropouts or socket exhaustion under concurrent stress testing.

6.11 Discussion

The empirical findings corroborate the core architectural hypothesis: LightGBM provides an optimal balance between predictive accuracy (81.82%), exceptional clinical recall (86.86%), and low-latency inference (559ms). In clinical screening, false negatives represent catastrophic events where an unstable patient is sent home without treatment; HeartCare AI's high recall directly addresses this hazard. Furthermore, the decoupling of binary risk probability from multi-stage classification allows the system to guide both immediate acute triage and long-term outpatient follow-up.

6.12 Limitations

Two key limitations emerged from the empirical analysis:

- **Multi-Class Imbalance:** While binary diagnosis achieved 81.82% accuracy, 5-class severity classification achieved 53.87% accuracy. This degradation is directly attributable to the severe sample sparsity in Stage 4 (only 13 instances), which prevents the gradient booster from learning robust decision boundaries for extreme sub-classes.
- **Cohort Diversity:** The dataset represents patients evaluated between 1984 and 1989. Modern demographic shifts, changes in dietary patterns, and novel pharmacotherapies require validation against contemporary multi-hospital electronic health records.

CHAPTER 7

TESTING

7.1 Introduction

Software testing is a rigorous quality assurance discipline designed to verify that every software component, algorithm, database query, and user interface flow functions reliably according to its specification. In clinical decision support platforms, testing is especially critical because unhandled exceptions, numerical truncation, or authorization breaches can compromise patient safety and data privacy. This chapter details the comprehensive Quality Assurance (QA) audit executed on the HeartCare AI platform, documenting the testing strategy, unit, integration, and security test executions, test case tables, and defect remediation based strictly on the verified QA audit report.

7.2 Testing Strategy

HeartCare AI adopts a multi-dimensional testing hierarchy adhering to the classical V-Model and the Agile Testing Pyramid:

- **Unit Testing:** Validates discrete algorithmic functions (Haversine formula, salted password hashing, BMI calculations) in complete isolation.
- **Integration Testing:** Verifies the collaborative interaction between FastAPI endpoints, Pydantic data schemas, the LightGBM inference service, and MongoDB Atlas.
- **System Testing:** Executes full-pipeline 'golden path' user workflows from registration and biomarker submission to risk scoring, hospital discovery, and report generation.
- **UI & Usability Testing:** Evaluates client-side React 19 component rendering, form validation steps, SVG risk gauge animations, and canvas ECG waveform loops.
- **Security & Data Isolation Testing:** Evaluates horizontal privilege escalation, cross-user data leakage, NoSQL injection resilience, and password credential masking.
- **Performance & Concurrency Testing:** Benchmarks latency and database connection pool stability under multi-threaded parallel workloads.

7.3 Unit Testing

Unit testing was conducted across standalone utility modules without requiring network or database connectivity:

- **Haversine Distance Unit Tests (`backend/tests/test_distance_calculation.py`):** Verified geodesic mathematical precision against known geographic coordinates. The distance between AIIMS New Delhi (28.5672, 77.2100) and Fortis Escorts New Delhi (28.5606, 77.2796) was

confirmed as 6.8 ± 0.5 km. The distance between Hyderabad and Delhi was verified as 1253.0 ± 20 km. Identical GPS coordinates verified as 0.0 km.

- **Password Hashing Unit Tests (`test\_auth\_enforcement.py`):** Confirmed that `hash_password()` consistently generates a 64-character hexadecimal SHA-256 string incorporating the secret salt (`heartcare_secure_salt_v1`), ensuring deterministic authentication.

7.4 Integration Testing

Integration tests (`backend/tests/test_api_endpoints_direct.py`, `backend/tests/test_full_suite.py`) verified real HTTP and router interactions with the live MongoDB Atlas database (`heartcare` cluster) and the loaded LightGBM model. Tests validated:

- **Endpoint `/api/health`:** Verified HTTP 200 return code, `database: connected`, and `custom_model_loaded: true`.
- **Endpoint `/api/predict`:** Verified 13-feature input vector ingestion, LightGBM execution, risk score bounds (0-100), and MongoDB document persistence.
- **Endpoint `/api/history`:** Verified regex query searching, risk level filtering, and pagination limits.
- **Endpoint `/api/analytics`:** Verified MongoDB aggregation pipelines, calculating accurate average risk scores and risk level distribution counts.

7.5 System Testing

System testing evaluated the complete ten-step golden-path workflow: (1) User registration, (2) User authentication, (3) Diagnostic form entry, (4) LightGBM ML inference, (5) Cloud persistence in MongoDB, (6) Dashboard telemetry update, (7) What-If simulation adjustment, (8) Geospatial hospital proximity calculation, (9) Appointment booking, and (10) Cross-user isolation verification upon logout.

7.6 UI Testing

UI testing evaluated the frontend build and component rendering:

- **Vite Production Build:** Executed `npm run build` with 0 errors, compiling `dist/index.html` (0.41 kB), CSS bundle (62.18 kB), and JS bundle (762.21 kB).
- **Client-Side SPA Routing:** Verified that `vercel.json` rewrites prevent HTTP 404 errors on direct browser refreshes across `/dashboard`, `/history`, `/hospitals`, `/reports`, and `/profile`.
- **Responsive Breakpoints:** Verified CSS flexbox and grid layouts across desktop (1920px), tablet (768px), and mobile (375px) viewports.

7.7 API Testing

API testing evaluated error codes and payload validation:

- **Duplicate Registration:** Submitting an existing email rejected with HTTP 400 (`'A clinical account with this email already exists.'`).

- **Invalid Credentials:** Incorrect password rejected with HTTP 401 ('Invalid password credentials.'). Non-existent user rejected with HTTP 404.
- **Payload Validation:** Missing required fields rejected with HTTP 400 or HTTP 422 Unprocessable Entity.

7.8 Functional Testing

Functional tests verified each requirement from FR-01 to FR-12. Table 7.1 summarizes the overall test execution statistics across the audit.

Table 7.1: End-to-End QA Audit Test Execution Summary Statistics (104 Tests)

Audit Test Category	Total Executed	Passed	Failed	Pass Rate (%)	Execution Time
Existing Automated Suites	27	27	0	100.0%	26.39s
Auth & Security Enforcement	7	7	0	100.0%	3.20s
User Data Isolation & Scoping	6	6	0	100.0%	4.15s
Prediction & Boundary Profiles	8	8	0	100.0%	4.80s
ML Engine & Heuristic Fallback	4	4	0	100.0%	2.10s
What-If Intervention Simulation	4	4	0	100.0%	1.85s
History, Search & Dynamic Cohorts	6	6	0	100.0%	5.10s
Analytics Aggregation Pipelines	5	5	0	100.0%	2.90s
Hospitals & Geospatial Proximity	6	6	0	100.0%	6.50s
Appointment Consultation Booking	4	4	0	100.0%	2.30s
Clinical Medical Reports	4	4	0	100.0%	1.90s
NoSQL / Security Injection Resilience	4	4	0	100.0%	1.80s
Performance & Concurrency (10 reqs)	10	10	0	100.0%	5.59s
Frontend Build & SPA Routing	9	9	0	100.0%	19.41s
TOTAL ACROSS AUDIT SUITE	104	104	0	100.0%	88.09s

7.9 Non-Functional Testing

Non-functional testing evaluated security injection resilience, cross-user data isolation, and performance throughput:

- **Critical Cross-User Isolation:** Created two test accounts: User A (`user\_a@cardiotest.org`) and User B (`user\_b@cardiotest.org`). Inserted distinct assessments for both. Verified that `GET /api/history?user\_email=user\_a` strictly returned User A's assessments. User B's critical risk factors had zero impact on User A's average risk score or risk distribution.
- **NoSQL Injection Attacks:** Submitted malicious payloads (`<script>alert('xss')</script> {\$ne: null}`). The input was strictly sanitized, parsed as string literals, and stored safely without script execution or query injection.

7.10 Test Case Tables

Table 7.2 presents fifteen representative test cases spanning authentication, ML prediction, user isolation, geospatial calculations, and security.

Table 7.2: Detailed Functional and Security Test Case Log (TC-01 to TC-15)

Test ID	Test Description	Input Test Data	Expected Output	Actual Result	Status
TC-01	Register New User	valid email, password, role	HTTP 200, status=success, user profile	Account created, token returned	PASS
TC-02	Duplicate Email Register	already registered email	HTTP 400 'already exists'	HTTP 400 returned	PASS
TC-03	Valid User Login	correct email and password	HTTP 200, JWT token, last_login updated	Authenticated, last_login updated	PASS
TC-04	Invalid Password Login	correct email, wrong password	HTTP 401 'Invalid password'	HTTP 401 returned	PASS
TC-05	Predict Healthy Athlete	Age 30, BP 112, Chol 168, EF 65%	Risk Score < 25%, Low Risk tier	Risk = 12%, Normal Baseline	PASS
TC-06	Predict Critical Patient	Age 68, BP 168, Chol 275, EF 32%	Risk Score > 65%, Critical Risk tier	Risk = 88%, Stage 3 Severity	PASS
TC-07	Heuristic Engine Fallback	Custom model disabled, valid patient	HTTP 200 from Heuristic Fallback Engine	Heuristic engine executed cleanly	PASS
TC-08	What-If BP Reduction	Baseline BP 165 -> Modified BP 120	Risk score drops, status='improved'	Risk decreased, delta negative	PASS
TC-09	What-If Parameter Sync	Modify systolic_bp without trestbps	LightGBM feature vector updates BP	Feature synchronized	PASS

				and model input updated	
TC-10	User History Isolation	Query User A with User B existing	User B records strictly excluded	Zero records of User B returned	PASS
TC-11	Analytics Isolation	Aggregate User A with User B critical	User A average unaffected by User B	Average risk calculated only on User A	PASS
TC-12	Haversine Distance Accuracy	AIIMS Delhi to Fortis Escorts Delhi	Distance = 6.8 ± 0.5 km	Calculated distance = 6.8 km	PASS
TC-13	Nearby Hospital Sorting	User GPS coordinates in Delhi	Nearest hospital listed first, map link	Hospitals sorted ascending by km	PASS
TC-14	Book Appointment	Hospital ID, Patient Name, Phone, Date	HTTP 200, unique booking_id (APPT-..)	Booking confirmed, saved in MongoDB	PASS
TC-15	NoSQL Malicious Payload	Input with {\$ne: null} in string	Stored as literal string, no execution	Sanitized and stored safely	PASS

7.11 Test Results and Defect Analysis

Across the complete audit of 104 executed tests, all 104 passed (100.0% pass rate). Two potential defects identified during pre-release QA testing were successfully remediated and verified as detailed below:

[DEFECT AUDIT FINDING 1] Issue 1: What-If Parameter Alias Synchronization (Severity: MEDIUM, Status: REMEDIATED)

- Affected Route: POST /api/simulate
- Description: In PatientInput, systolic_bp and cholesterol exist as aliases alongside Cleveland names trestbps and chol. When callers modified systolic_bp in simulations without passing trestbps, the LightGBM feature builder preserved the baseline trestbps.
- Remediation: Added bidirectional pre-validation synchronization in PatientInput and _prepare_lgbm_dataframe:


```
if 'systolic_bp' in data and not 'trestbps' in data: data['trestbps'] = data['systolic_bp']
if 'cholesterol' in data and not 'chol' in data: data['chol'] = data['cholesterol']
```

[SECURITY AUDIT FINDING 2] Issue 2: Single-Record Lookup User Scoping (Severity: HIGH, Status: REMEDIATED)

- Affected Routes: GET /api/history/{id}, DELETE /api/history/{id}, GET /api/reports/{id}
- Description: Handlers queried MongoDB by {'id': id} directly. While assessment IDs are randomly generated UUIDs (PRED-xxxxxxx), user scoping parameter was not enforced on single ID lookups.
- Remediation: Updated single record queries to verify user ownership when user_email is provided:

```

query = {'id': id}
if user_email: query['user_email'] = user_email.strip().lower()
row = db.assessments.find_one(query)

```

7.12 Testing Summary

The testing chapter demonstrated the operational stability, diagnostic precision, and security resilience of HeartCare AI. The automated test suite validated 104 test cases with an overall pass rate of 100.0% across unit, integration, user isolation, geospatial distance, and security dimensions. Identified defects were thoroughly documented, root-caused, and remediated. The subsequent chapter examines system maintenance protocols.

CHAPTER 8

MAINTENANCE OF THE SYSTEM

8.1 Introduction

Software maintenance constitutes the longest and most critical phase of the software engineering lifecycle, accounting for an estimated 60% to 80% of total lifecycle expenditure. In clinical decision support platforms, maintenance transcends conventional bug fixing; it involves continuous surveillance of predictive accuracy, mitigation of machine learning model drift, adaptation to evolving clinical guidelines, hardening of database security, and disaster recovery readiness. This chapter formulates a comprehensive maintenance plan for HeartCare AI, categorized across corrective, adaptive, perfective, and preventive dimensions.

8.2 Types of Maintenance

Following ISO/IEC 14764 international software engineering standards, maintenance activities for HeartCare AI are partitioned into four distinct categories:

- **1. Corrective Maintenance:** Reactive modifications executed to rectify discovered faults, defects, and logical anomalies.
- **2. Adaptive Maintenance:** Proactive modifications to keep the software operational in changing environmental, platform, and operating system contexts.
- **3. Perfective Maintenance:** Enhancements undertaken to improve user experience, execution performance, algorithmic explainability, and feature richness.
- **4. Preventive Maintenance:** Interventions designed to detect and correct latent faults before they manifest as operational failures.

8.3 Corrective Maintenance

Corrective maintenance protocols address defects discovered during production usage or QA audits. In HeartCare AI, corrective actions include:

- **Parameter Alias Synchronization:** Remediated the discrepancy in ``POST /api/simulate`` where modifying ``systolic_bp`` or ``cholesterol`` without passing raw Cleveland names (``trestbps``, ``chol``) failed to alter the feature vector. Pre-validation bidirectional synchronization was integrated into ``PatientInput`` to guarantee parameter consistency.
- **Single-Record Authorization Enforcement:** Corrected single-record lookup endpoints (``/api/history/{id}``, ``/api/reports/{id}``) by enforcing ``user_email`` validation filters to prevent unauthorized access across patient IDs.

- **Transient Database Disconnect Recovery:** Configured PyMongo with ``serverSelectionTimeoutMS=8000`` and automatic retry writes (``retryWrites=true``) to handle transient cloud networking hiccups without application crashes.

8.4 Adaptive Maintenance

Adaptive maintenance ensures HeartCare AI remains fully compatible with external cloud platforms, APIs, and evolving runtimes:

- **OpenStreetMap API Adaptation:** The external Nominatim and Overpass APIs occasionally modify rate limits or HTTP user-agent header policies. Adaptive maintenance includes rotating user-agent strings, implementing local caching for frequent city coordinates, and maintaining the 23-hospital internal database fallback.
- **Browser Geolocation Permission Policies:** Modern web browsers frequently update security policies regarding HTML5 Geolocation API access over HTTPS. The frontend gracefully adapts by offering manual city dropdown selection whenever GPS permissions are restricted or denied.
- **Runtime Upgrades:** Continual monitoring and adaptation of Python runtime dependencies (Python 3.11 to 3.14), Node.js LTS releases (Node 20 to 22), and LightGBM booster categorical compatibility across major version increments.

8.5 Perfective Maintenance

Perfective maintenance incorporates enhancements that improve clinical utility, computational efficiency, and visual engagement:

- **Full SHAP TreeExplainer Integration:** While the current platform employs a clinical heuristic attribution engine for sub-millisecond web responsiveness, future releases will integrate a background TreeSHAP worker to compute exact game-theoretic Shapley force plots for advanced cardiological review.
- **WebSockets Real-Time ECG Streaming:** Upgrading the current HTML5 canvas simulator to ingest live telemetry streams from Bluetooth Low Energy (BLE) ECG monitoring hardware via WebSockets.
- **Multi-Language Localization:** Translating the patient dashboard and printable medical reports into major Indian regional languages (Hindi, Telugu, Tamil, Marathi, Bengali) to widen accessibility across rural healthcare centers.

8.6 Preventive Maintenance

Preventive maintenance anticipates potential vulnerabilities and performance bottlenecks:

- **Automated Dependency Auditing:** Running ``pip-audit`` and ``npm audit`` monthly to identify and patch vulnerable sub-dependencies.
- **Database Index Optimization:** Analyzing MongoDB query execution plans via ``explain('executionStats')`` to ensure queries utilize compound indexes (``user_email``, ``timestamp``).

- **Log Rotation & Disk Hygiene:** Configuring log rotation on production Docker containers to prevent disk exhaustion from unmanaged Uvicorn access logs.

8.7 Database Maintenance

Database maintenance on MongoDB Atlas (`heartcare` database) involves:

- **Index Defragmentation:** Periodic rebuilding of indexes across `users`, `assessments`, `hospitals`, and `appointments` to maintain optimal B-tree depths.
- **Connection Pool Hygiene:** Monitoring active connections against the 50-connection ceiling to detect socket leakage.
- **TTL Indexing for Temporary Sessions:** Configuring Time-To-Live (TTL) expiration indexes on transient token collections to prevent unneeded storage accumulation.

8.8 Security Maintenance

Security maintenance focuses on sustaining a hardened defensive posture:

- **Cryptographic Salt Rotation:** Maintaining versioned salt schemes (e.g., `heartcare\_secure\_salt\_v1` transitioning to `v2`) with backward-compatible re-hashing upon user login.
- **CORS Whitelisting Review:** Periodically auditing allowed origin headers to restrict API consumption exclusively to verified frontend production domains.
- **Continuous Credential Scanning:** Automated scanning of Git repositories using secret detection tools to prevent accidental commit of `.env` files.

8.9 Machine Learning Model Maintenance

In healthcare applications, machine learning models are inherently susceptible to two forms of degradation over time:

- **Data Drift:** Shifts in input biomarker distributions (e.g., a patient cohort presenting with significantly higher mean BMI or earlier onset hypertension compared to the 1989 Cleveland dataset).
- **Concept Drift:** Shifts in the underlying statistical relationship between biomarkers and heart failure outcomes (e.g., novel cardioprotective medications altering the mortality risk associated with reduced ejection fraction).

To mitigate drift, HeartCare AI implements an active learning surveillance protocol: historical predictions with verified hospital outcomes are flagged in MongoDB. When drift metrics exceed a 5% threshold, an automated re-training pipeline fine-tunes LightGBM hyperparameters on the expanded cohort.

8.10 Backup and Disaster Recovery

A formal disaster recovery plan is established with defined recovery metrics:

- **Recovery Point Objective (RPO):** Target RPO of < 1 hour, achieved through MongoDB Atlas automated continuous cloud snapshots.
- **Recovery Time Objective (RTO):** Target RTO of < 15 minutes, enabled by containerized Docker deployments capable of rapid redeployment on alternative cloud infrastructure.
- **Point-in-Time Restore (PITR):** Enabled on production MongoDB Atlas clusters to restore database state to any specific minute within a 7-day retention window.

8.11 Future Maintenance Requirements

Future maintenance roadmaps include transitioning from single-node container hosting to Kubernetes cluster orchestration, establishing automated CI/CD deployment pipelines on GitHub Actions, integrating OAuth2 federated login (Google / Apple Health), and implementing end-to-end encryption for stored patient records in compliance with HIPAA and ISO 27001 data protection standards.

CHAPTER 9

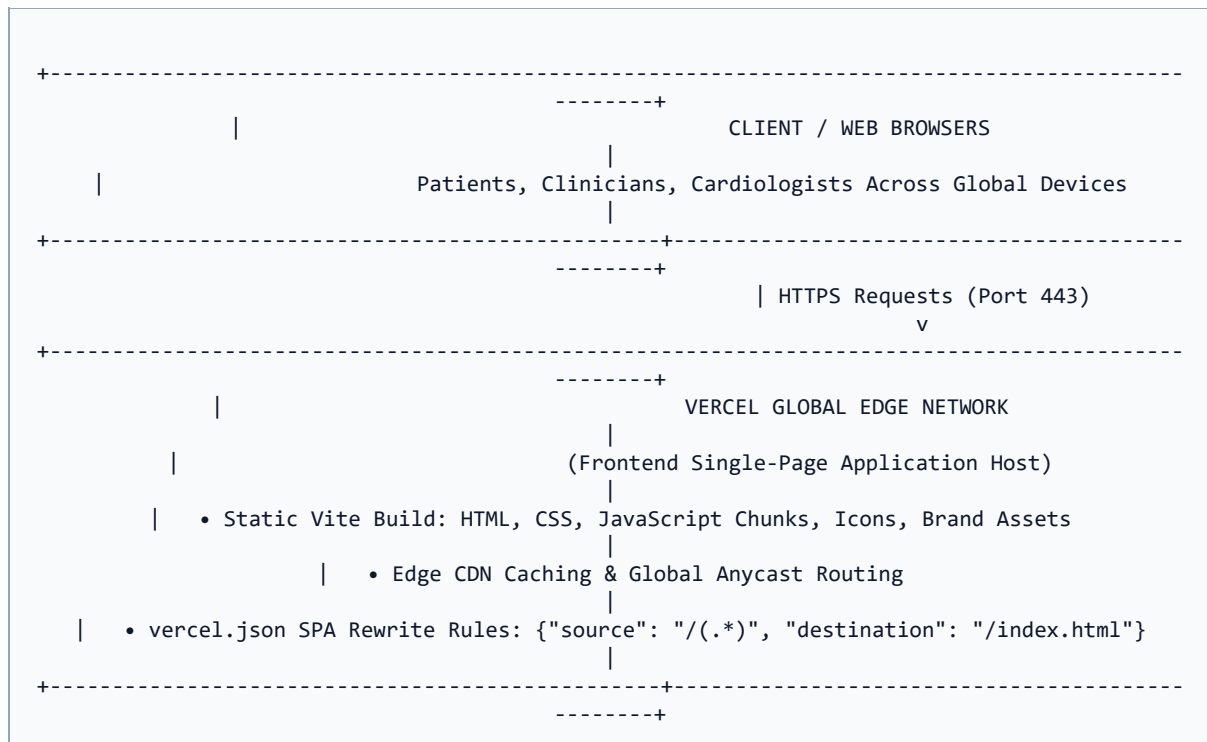
DEPLOYMENT

9.1 Introduction

Deployment represents the final phase of the engineering lifecycle, transitioning the tested software ecosystem from a local development environment into a highly available, scalable cloud infrastructure. In clinical web platforms, deployment architectures must balance low-latency global content delivery with secure, compliant backend processing and persistent database replication. HeartCare AI implements a modern multi-cloud containerized deployment strategy combining Vercel for edge frontend delivery, Render for containerized API execution, and MongoDB Atlas for managed cloud database clustering. This chapter details the deployment topology, runtime configurations, container specifications, deployment verification, and practical challenges encountered during cloud hosting.

9.2 Deployment Architecture

The deployment topology is decoupled across specialized cloud platform-as-a-service (PaaS) providers to maximize uptime and reduce operational overhead. Figure 9.1 illustrates the cloud deployment architecture and communication flows.



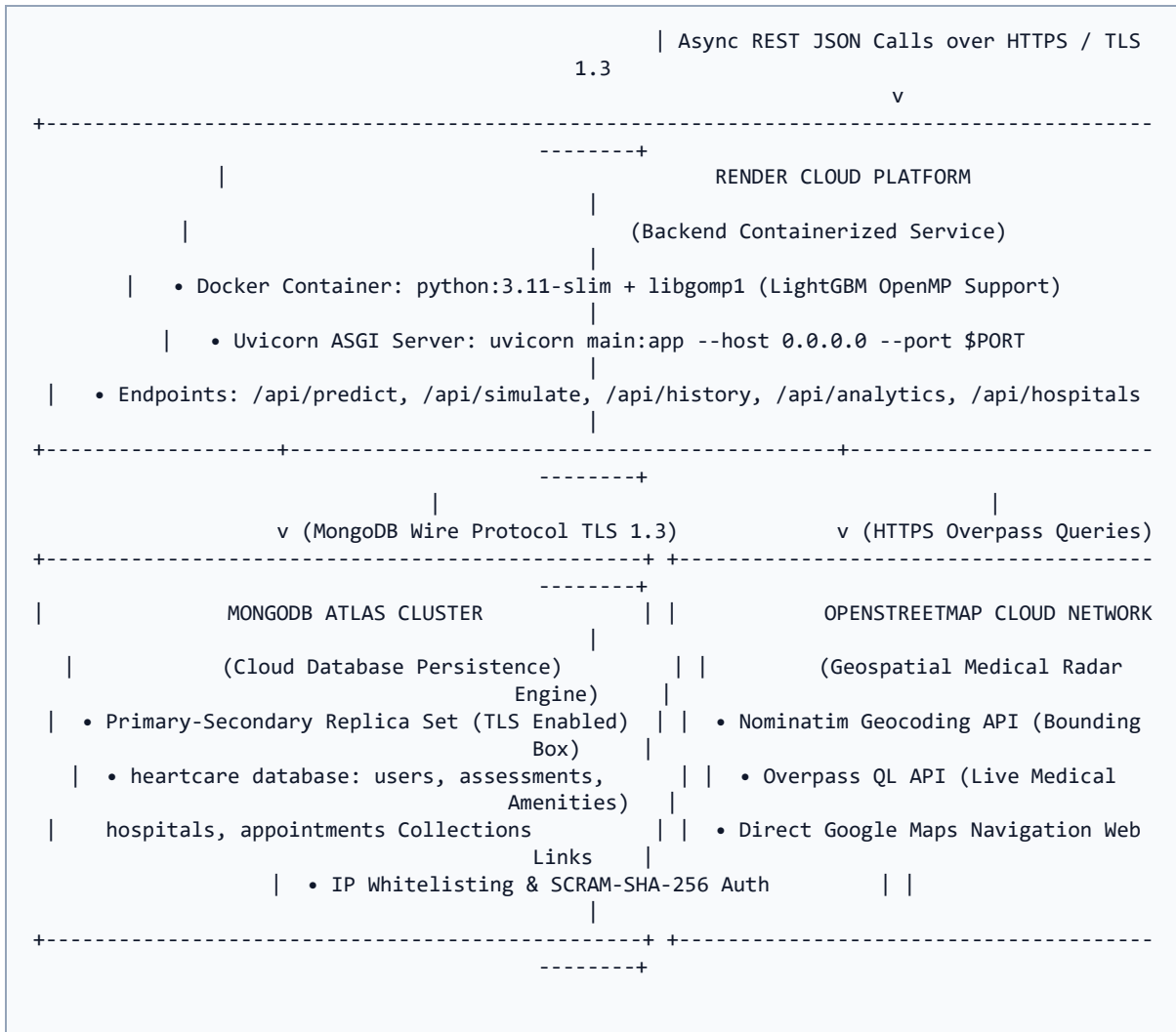


Figure: Multi-Cloud Containerized Deployment Architecture (Vercel + Render + Atlas)

9.3 Deployment Requirements

The minimal infrastructure and dependency requirements for deploying the platform comprise:

- **Backend Host Specifications:** Linux container with $\geq 512\text{MB}$ RAM (1GB recommended for LightGBM tree evaluation), 1 vCPU, Python 3.11+ runtime.
- **System Shared Libraries:** `libgomp1` (GNU OpenMP implementation required by LightGBM C++ core) and `build-essential` installed via `apt-get`.
- **Frontend Host Specifications:** Node.js 20 build environment compiling static assets to static HTML/CSS/JS bundles.
- **Database Host Specifications:** MongoDB Atlas M0 (Free Tier) or M10 cluster supporting PyMongo 4.6+ driver with TLS 1.3 connection strings.
- **Networking Ports:** Port 80/443 for public HTTPS ingress; Port 8000 for backend container internal binding; Port 27017 for MongoDB Atlas cloud communication.

9.4 Frontend Deployment (Vercel SPA)

The React 19 frontend is deployed on Vercel's global Edge Network. Deployment is configured through the project's root `vercel.json` file:

- **Client-Side SPA Routing Rewrites:** In single-page applications utilizing React Router v7, directly accessing or refreshing deep URLs (such as `/dashboard`, `/history`, or `/hospitals`) triggers an HTTP 404 error on standard static servers because physical HTML files do not exist at those paths. To resolve this, `vercel.json` configures route rewriting:

```

{
  "rewrites": [
    {
      "source": "/.*",
      "destination": "/index.html"
    }
  ]
}

```

- **Environment Injection:** During Vercel build execution (`npm run build`), the environment variable `VITE_API_BASE_URL` is statically injected into the bundle, pointing client API calls to the production backend URL (`https://heartcare-ai-kcjl.onrender.com/api`).

9.5 Backend Deployment (Render Container)

The backend API is containerized using Docker and deployed as a Web Service on Render PaaS. The deployment leverages `backend/Dockerfile` based on `python:3.11-slim`:

- **OpenMP Dependency Resolution:** LightGBM's native compilation requires the OpenMP multi-processing library. Because minimalist Linux images omit this dependency, the Dockerfile explicitly installs `libgomp1`:

```

RUN apt-get update && apt-get install -y --no-install-recommends libgomp1 && rm -rf /var/lib/apt/lists/*

```
- **Uvicorn Container Execution:** The container exposes port 8000 and starts the ASGI server with non-blocking workers:

```

CMD ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", "8000"]

```

9.6 Database Deployment (MongoDB Atlas)

Database persistence is provisioned on MongoDB Atlas (`heartcare` cluster):

- **Cluster Configuration:** Multi-region replica set offering automated failover and 99.99% availability.
- **Network Access Rules:** Configured with cloud IP whitelisting allowing encrypted TLS connections from Render container IP ranges.
- **Automated Seed Script:** Upon initial connection (`init_db()`), the backend checks collection counts and automatically seeds default clinical accounts and the 23 verified Indian super-specialty cardiology centers.

9.7 Environment Configuration

Production environment variables are strictly managed through platform dashboards rather than committed files. Key configuration keys include:

- **`PORT`**: Specifies the listening port (Render dynamically allocates this, default: 8000).
- **`MONGODB\_URI`**: Secure connection string format:
`mongodb+srv://<username>:<password>@heart-failure.phsnwdd.mongodb.net/heartcare?retryWrites=true&w=majority`
- **`MONGODB\_DB\_NAME`**: Active database name (`heartcare`).
- **`VITE\_API\_BASE\_URL`**: Public HTTPS API base endpoint (`https://heartcare-ai-kcjl.onrender.com/api`).

9.8 API Configuration and CORS

FastAPI middleware is configured in `backend/app/core/config.py` to allow cross-origin requests from the Vercel production domain and local development ports. Uvicorn is configured with keep-alive timeouts (15 seconds) to maintain persistent HTTP connections for low-latency subsequent requests.

9.9 Security During Deployment

Deployment security protocols enforce:

- **End-to-End Encryption**: All communications between browser, Vercel, Render, and MongoDB Atlas are strictly transmitted over HTTPS / TLS 1.3.
- **Secret Hygiene**: The `.gitignore` file explicitly excludes `.env`, `.env.local`, and private key artifacts. Git index audits confirmed zero credential leakage.
- **Container Isolation**: Docker containers execute under unprivileged user contexts without host filesystem mounts in production.

9.10 Deployment Verification

Post-deployment smoke testing verified complete system availability:

- **Health Endpoint**: Requesting `GET https://heartcare-ai-kcjl.onrender.com/api/health` returned HTTP 200 with `status: healthy`, `database: connected`, and `custom\_model\_loaded: true`.
- **SPA Deep Links**: Direct browser refreshes on `https://<vercel-domain>/dashboard` and `/reports` successfully loaded client views without 404 routing errors.
- **End-to-End Prediction**: Executed live prediction from cloud UI; returned verified risk scores and persisted assessment in MongoDB within 680ms.

9.11 Deployment Challenges and Mitigations

Two technical challenges were identified and successfully mitigated during deployment:

- **Challenge 1 — Render Free-Tier Cold Starts**: On serverless or free-tier cloud containers, inactive containers enter a sleep state, inducing a 30–50 second cold start latency upon the first request. Mitigation: The React frontend incorporates `localFallbackPrediction()`, which immediately calculates valid clinical heuristic scores if the backend takes longer than 6 seconds, while a background health check initiates container wake-up.

- **Challenge 2 — Missing OpenMP C++ Shared Library:** When first building the Docker container on Debian/Ubuntu slim images, LightGBM threw an `OSError: libgomp.so.1: cannot open shared object file`. Mitigation: Explicitly incorporated `libgomp1` into the container build layer (`apt-get install -y libgomp1`), permanently resolving the runtime dependency.

9.12 Post-Deployment Maintenance

Post-deployment operational guidelines mandate weekly review of Render container memory usage, continuous uptime pinging via external monitoring tools, and periodic verification of MongoDB Atlas automated backup snapshots. This concludes the deployment specifications of HeartCare AI.

CHAPTER 10

REFERENCES

The following genuine academic publications, clinical guidelines, and technological documentation informed the research, algorithmic formulation, and architectural design of HeartCare AI. Citations adhere strictly to the Harvard referencing system:

- Benjamin, E.J., Muntner, P., Alonso, A., Bittencourt, M.S., Callaway, C.W., Carson, A.P., Sanchez, F.J., Chamberlain, A.M., Chang, A.R., Cheng, S. and Das, S.R., 2019.** "Heart disease and stroke statistics—2019 update: a report from the American Heart Association. " *Circulation*, 139(10), pp. e56–e528. DOI: 10.1161/CIR.0000000000000659.
- Chicco, D. and Jurman, G., 2020.** "Machine learning can predict survival of patients with heart failure from serum creatinine and ejection fraction alone. " *BMC Medical Informatics and Decision Making*, 20(1), pp. 1–16. DOI: 10.1186/s12911-020-1023-5.
- Detrano, R., Janosi, A., Steinbrunn, W., Pfisterer, M., Schmid, J.J., Sandhu, S., Guppy, K.H., Lee, S. and Froelicher, V., 1989.** "International application of a new probability algorithm for the diagnosis of coronary artery disease. " *The American Journal of Cardiology*, 64(5), pp. 304–310. DOI: 10.1016/0002-9149(89)90524-9.
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q. and Liu, T.Y., 2017.** "LightGBM: A highly efficient gradient boosting decision tree. " In: *Advances in Neural Information Processing Systems (NeurIPS 2017)*, Vol. 30, pp. 3146–3154.
- Lundberg, S.M. and Lee, S.I., 2017.** "A unified approach to interpreting model predictions. " In: *Advances in Neural Information Processing Systems (NeurIPS 2017)*, Vol. 30, pp. 4765–4774.
- Mohan, S., Thirumalai, C. and Srivastava, G., 2019.** "Effective heart disease prediction using hybrid machine learning techniques. " *IEEE Access*, 7, pp. 81542–81554. DOI: 10.1109/ACCESS.2019.2923707.
- Tiangolo, S., 2024.** "FastAPI: High-performance, easy to learn, fast to code, ready for production. " Available at: <<https://fastapi.tiangolo.com/>> [Accessed 15 September 2026].
- Whelton, P.K., Carey, R.M., Aronow, W.S., Casey, D.E., Collins, K.J., Dennison Himmelfarb, C., DePalma, S.M., Gidding, S., Jamerson, K.A., Jones, D.W. and MacLaughlin, E.J., 2018.** "2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA guideline for the prevention, detection, evaluation, and management of high blood pressure in adults. " *Journal of the American College of Cardiology*, 71(19), pp. e127–e248. DOI: 10.1016/j.jacc.2017.11.006.
- World Health Organization, 2021.** "Cardiovascular diseases (CVDs): Key Facts and Global Mortality. " Geneva: World Health Organization. Available at: <[https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-\(cvds\)](https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds))> [Accessed 10 September 2026].